

Differentiation of physical and virtual real estate in the application of tokenization

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Abstract

In recent years, blockchain technology has established itself as one of the most important innovations. This technology offers enormous potential and enables existing industries to evolve. One of the relevant applications of blockchain technology is in the real estate industry. Tokenization as a concept based on blockchain technology has the potential to address and solve existing problems in the real estate market. However, tokenization goes one step further and enables virtual real estate to be owned and traded in digital environments. The virtual real estate that resides in these digital worlds finds similarities in its nature. Nevertheless, there are a lot of other differences as well, especially when the focus is on the challenges and potentials of tokenization of these two types of real estate which are analyzed and considered. The goal of this research is to identify these differences between physical and virtual real estate in terms of tokenization. With a qualitative approach in the form of an expert interview with seven professionals, the differences were determined. Although tokenization of virtual real estate is a more recent phenomenon, it is already more developed than tokenization of physical real estate. However, the potential of the physical variant is estimated to be much higher. In addition to their challenges and potential, they differ in their nature.

Keywords: Blockchain; Tokenization; Real estate; Metaverse

1 Introduction

The first chapter of this thesis is divided into four sections. The first describes the fundamental motivation of the research. The second concerns previous studies, and it identifies the research gap. The objectives of the thesis and the research questions are defined in the third section. The fourth section contains a brief overview of the structure of the work.

1.1 Motivation

Due to its profitability and its capacity to provide consistent revenue through rentals and leases, real estate is one of the most dependable assets for investors (Gupta et al., 2020, p. 77). Beyond its economic significance, real estate is also important socially because housing satisfies the basic human need for shelter. Therefore, housing possesses a special value (A'zamovna & Hamrokulovna, 2021, p. 86). Pension funds, life insurance companies, sovereign wealth funds, and other institutional investors treat commercial real estate as a significant asset class in their portfolios. Commercial real estate is another investment option for wealthy individuals. Commercial real estate transactions occur in private illiquid marketplaces, not in the public markets in which other assets are traded, such as common stock exchanges (Ghent et al., 2019, p. 1). Real estate, historically one of the most conservative markets, has undergone significant changes as a result of the shifts in the global economy since the early 1990s, including the globalization of financial and commodity markets, the deregulation of the financial sector, and the expansion of the role of finance in the economy. Processes that are linked to disruptive technologies have caused this tendency to become more pronounced over the last decade (Sternik & Safronova, 2021, p. 676).

Blockchain is one of these disruptive technologies. It has become popular mainly because of cryptocurrencies. However, blockchain has a wide range of alternative applications and therefore possesses enormous potential (Treleaven et al., 2017, p. 17). Consequently, much attention has been paid to developments such as smart contracts and tokenization on the blockchain. Although governments and industries have begun to focus on it, the technology has not reached its full potential yet. Given the possibility of automatic confirmation and the transparent character of transactions, the real estate industry is among the sectors that could benefit substantially from distributed ledger technology (DLT; Chapman et al., 2019, p. 1). At present, the real estate market confronts significant issues, such as information inefficiencies and high transaction costs, because it is essentially heterogeneous and illiquid. These challenges can also be attributed to a lack of market transparency (Wouda & Opdenakker, 2019, pp. 570-572). Furthermore, the need for intermediaries, such as banks, and the repeated verification of documents during transfers of ownership increase costs considerably (Kalyuzhnova, 2018, p. 5). Lack of liquidity is also considered to be a disadvantage of real estate.

Blockchain technology provides specific solutions to these issues. Tokenization, which is made possible by the blockchain, enables the problems of liquidity and transparency to be solved. Furthermore, the market can attract small investors. Fractional ownership, which enables investors to purchase real estate shares in the form of tokens and to swap them on secondary marketplaces, is a crucial means of generating liquidity (A. Gupta et al., 2020, p. 77). Tokenization thus allows a physical asset to be represented digitally on a blockchain in the form of a token.

The promise of tokenization extends far beyond these possibilities. Almost everything, be it a physical asset, a service, or a voting right, can be tokenized. A token can also represent digital goods as non-

fungible tokens (NFTs) that exist only in online environments (Sazandrishvili, 2020, p. 68). Digital art, as well as any other special digital property for which ownership may be documented through smart contracts on a blockchain, can be represented by NFTs. NFTs are currently the subject of intense hype (Pinto-Gutiérrez et al., 2022, p. 1). The hype that surrounds NFTs in art has been pushed into a new direction, specifically that of virtual real estate, by the metaverse. Many new projects have been established that aim to create a virtual world, which has given real estate a new meaning. The decentralized Sandbox world is one of them. In that world, virtual real estate can be exchanged through NFTs (Macedo et al., 2022, pp. 13-16).

The author recognizes that blockchain is still in its infancy and that its investigation is far from complete. It has proven to be one of the most revolutionary technologies since the emergence of the internet. There is more substance to it than just hype and cryptocurrencies. Tokenization, in particular, is considered to be one of the key applications of blockchain technology. Almost anything can be tokenized. Tokenization is seen as particularly important in the context of real estate. Real estate is a one-of-a-kind asset. For decades, the traditional real estate market has mostly remained unchanged. Physical real estate tokenization is still in its early stages and requires more investigation. It is especially important to comprehend the recent emergence of the metaverse, which elevates real estate to a new and virtual level. Virtual real estate, when combined with tokenization, is a unique and even more recent phenomenon that has been subjected to almost no research. Finally, the goal of this study is to present physical and virtual real estate in terms of tokenization and to provide an overview of the concepts. Its objective is to establish a theoretical foundation and to raise awareness of the main issues in the field.

1.2 Previous studies and research gap

Blockchain has made many headlines in recent years. It was introduced by Satoshi Nakamoto in 2008 as a foundation technology for the first global decentralized cryptography-based digital currency, Bitcoin. The optimal means of applying blockchain are now being discussed in nearly every field. As a result, many believe that it can change the world (Panwar & Bhatnagar, 2020, p. 1). Tokenization is one of the applications of blockchain technology. Tokenization transfers information, values, and rights that are associated with an asset to the blockchain. As a result, tokens, like traditional securities, can be traded on secondary markets. From this perspective, tokenization can make the financial industry more accessible, cheaper, faster, and easier to use because illiquid assets can be unlocked and because trade volume can increase considerably (Tian, Lu, et al., 2020, p. 489).

The concept of tokenization has piqued the curiosity of the public. Few individuals comprehend it. The subject is still relatively new and requires further investigation. Much progress needs to be made before the public can understand and accept this approach. For general acceptance to be conceivable, a large number of users must begin using blockchain tokenization. Most tokenization companies have existed for less than five years. Furthermore, the majority of the businesses that are involved in the process are still in their early stages of development and are classified as start-ups (Tian, Lu, et al., 2020, pp. 495-496).

Blockchain technology has the potential to alter business processes and industries. Blockchain technology, which entails tokenization, is still in its early stages, particularly in its application in the real estate business. Many use cases are being identified at present through continual experimentation. While many theories show how the technology can improve many elements of the market, reality is usually different (Thota, 2019, p. 56). Due to the recent growth in the importance and potential of blockchain technology,

including tokenization, in real estate, a research gap has emerged—many studies present the theoretical benefits of the technology for the real estate industry without referring to the current state of affairs.

In addition, a completely new development has emerged. Financial and investment organizations across the globe are interested in the token economy and in Web 3.0. The metaverse concept of the 1990s has lately been revitalized. It refers to a parallel virtual world that is always live and synchronized. The open blockchain enables the processing of NFT transactions in a secure and verifiable manner. Within this metaverse, real estate can be acquired, and ownership can be transferred by using NFTs (Chuen & Won, 2022, pp. 1-3). The NFT market is fairly young and developing rapidly, especially when NFTs and virtual real estate are combined (Dowling, 2021, p. 4). Few authors have investigated the phenomenon of metaverse-based virtual real estate. Yoo (2022) was one of the first to contribute to the field by researching virtual-space real estate services (Yoo, 2022, p. 83). Dowling (2021) also wrote on the metaverse, conducting preliminary research on virtual real estate and examining the pricing of plots in the largest virtual blockchain world, which is called Decentraland. Thus, his contribution is one of the first in the field (Dowling, 2021, p. 1). In this context, a further research gap becomes apparent. The tokenization of virtual real estate, which is recognized as a form of NFT, is significantly less well researched than the tokenization of physical real estate. However, this market is becoming increasingly important, necessitating additional research.

1.3 Research question and objectives

The purpose of this thesis is to examine the tokenization of physical and virtual real estate in order to plug the research gaps that were highlighted in Section 1.2. Both subjects require further investigation. The thesis further compares physical and virtual real estate, a research gap that has not been explored previously. As a result, the author has formulated the main *research question* as follows:

“What is the difference, in terms of tokenization, between physical real estate and virtual real estate?”

Objectives:

1. To draw a distinction between the two variables in this study, namely physical real estate and virtual real estate, by isolating the theoretical basis of the elements of tokenization for each and by describing current events in a manner that can raise awareness and generate foundational knowledge;
2. To identify potential areas for the future development of the tokenization of physical and virtual assets, such as real estate, in line with the exigencies of the actual market situation;
3. To address challenges in the current development of the tokenization of physical and virtual assets, such as real estate, from the perspective of industry participants with years of experience in the field.

1.4 Thesis structure

The thesis is organized into six chapters. The chapters are divided into sections for ease of exposition. Each chapter serves a distinct purpose. The purpose of the first chapter is to introduce the reader to the author's motivation for conducting research in the field as well as to present current research on the tokenization of physical and virtual real estate. In this way, the research question and the goals of the work are defined.

The second chapter examines the three main components of the topic and the current state of research. It contains multiple sections. The author first examines the theoretical foundations and the fundamental features of blockchain technology. Thereafter, tokenization is discussed in relation to both physical and virtual real estate.

The methodology that was adopted to answer the underlying research questions is described in depth in Chapter 3. First, the focus is placed on the literature review in Chapter 2. Thereafter, the exposition turns to the qualitative expert interviews. The expert interview is described as a special form of guided interview. In addition, the research procedure, from establishing guidelines to conducting interviews, is described. An additional chapter is dedicated to expert selection. Finally, the concept of content analysis is discussed in relation to the analysis of the data.

The results are presented in Chapter 4. Those results are evaluated by using the procedure that is described in Section 3.7. The empirical research findings thus serve as a foundation for the discussion.

The theoretical and empirical findings are discussed in the penultimate chapter. Physical real estate is contrasted with virtual real estate in terms of tokenization, which allows the two to be differentiated. The main goal of that chapter is to answer the research question.

Finally, the conclusion in Chapter 6 highlights and summarizes the main findings. Furthermore, the chapter identifies potential research directions for the future as well as the limitations of the research.

2 Literature Review

This chapter is intended to provide the reader with a fundamental understanding of the most significant aspects of the research topic. The main topic of this research work is tokenization in the markets for physical and virtual real estate. Therefore, an overview of the underlying blockchain technology is presented first. In addition, the chapter discusses important features of the blockchain, such as smart contracts and tokenization in general. These features are essential for understanding the main topic. Subsequently, tokenization is discussed in detail in relation to physical real estate and virtual real estate. Furthermore, the phenomena of NFTs and the metaverse are examined insofar as they are related to virtual real estate. This theoretical basis makes it possible to understand the research topic and to compare physical real estate and virtual real estate in the final part of the master's thesis.

2.1 Theoretical background: Blockchain

This section concerns the technical aspects of blockchain and DLT. Furthermore, it covers the fundamentals of blockchain technology and enables the reader to recognize and understand connections in the subsequent chapters. Therefore, the section is indispensable as an introduction to the terminology of tokenization, which is connected to physical and virtual real estate.

In the world of internet of things, social media, and cloud computing, blockchain is the latest addition to the technology. The swift development of blockchain and its respected applications has made it a force to be recognized. But the scope of blockchain is not limited to Bitcoin or the cryptocurrencies implementation (Memon et al., 2018, p. 29).

As the quote from Memon et al (2018) above shows, blockchain is more than just the technology behind the cryptocurrency Bitcoin. Blockchain technology is a real innovation that has considerable potential and many applications (Alladi et al., 2019, p. 2).

Blockchain technology was first outlined in a white paper called "Bitcoin: A Peer-to-Peer Electronic Cash System," which was written by the pseudonymous Satoshi Nakamoto in 2008. Nakamoto's white papers, the identity of whose author remains a mystery to this date, were not published in a peer-reviewed journal. However, the cryptocurrency Bitcoin, which uses blockchain technology, laid the foundation for other applications. The intention and the goal of the white paper is relatively clear. Blockchain provides a decentralized and cryptographically secure system. Therefore, the implementation of the blockchain within Bitcoin made it possible to send online payments directly from one party to another without the need for a trusted authority or a financial institution (Nakamoto, 2008).

2.1.1 Definition

Blockchain technology is often used in connection with the cryptocurrencies Bitcoin and Ethereum. However, blockchain has many other applications. Thus, blockchain technology exists in various forms (Tasatanattakool & Techapanupreeda, 2018, p. 473). In essence, blockchain technology is a database of transactions records that is distributed, validated, and maintained by a network of computers around the globe. No single central authority, such as a bank, is needed. All records are supervised by a large community and, therefore, no individual can change or delete transaction history (Sarmah, 2018, p. 23). It seems that blockchain promises a bright future for humanity. It solves a variety of problems, such as trust, transparency, security, and the reliability of data processing (Golosova & Romanovs, 2018, p. 1). Additionally, when blockchain technology is used, information cannot be manipulated. The same is not

true of conventional centralized databases. The distributed structure of blockchain and the guarantees that peers in the network offer ensure that this advantage can be secured. A normal centralized database is typically located on an individual server, whereas blockchain is distributed between the users of the software and therefore decentralized. It is possible for everyone on the blockchain network to access all entries. Therefore, no central entity can seize control of the network. The transaction is transmitted to the network. As a result, computer algorithms determine its authenticity (Sarmah, 2018, p. 23). After the transaction is completed successfully, it is linked and connected to previous transactions. A chain of transactions is thus formed. In other words, each block is connected to the previous block by a cryptographic hash. This feature of the process is the main reason why it is called “blockchain.” A block can be defined as a data structure that allows a list of transactions to be stored (Wust & Gervais, 2018, p. 45). The concept is illustrated in Figure 1 below. Each block comprises a sequence of transactions. Cryptographic hashes also ensure that prior blocks cannot be altered. If a prior block is changed, the new hash would not match the previously recorded one, and the link between the two blocks would be broken (Xu et al., 2019, p. 5).

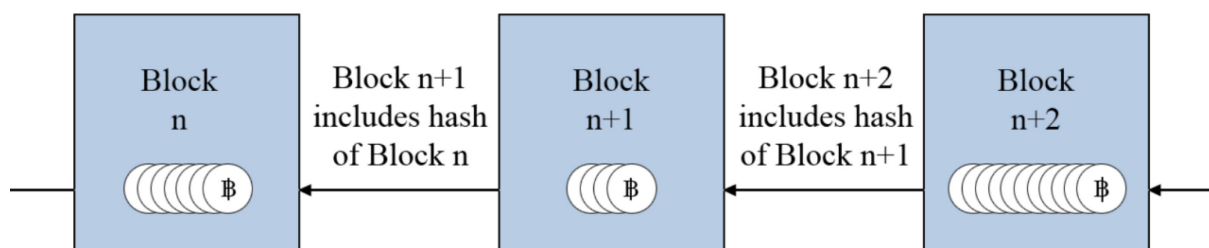


Figure 1: Blockchain data structure

Source: Xu et al., (2019, p. 5)

Blockchain technology is obviously based on a decentralized network, which means that it functions as a peer-to-peer network. Bitcoin, as mentioned previously, is one of the most famous blockchain technologies. It hosts a digital ledger. In Bitcoin, a complex computer algorithm, which is attached to a distributed network, makes it possible to mine, store, and trade bitcoins. Furthermore, the application of blockchain is not limited to transactions; it can also be used as registry and inventory for all types of assets (Sarmah, 2018, p. 23).

2.1.2 Distributed ledger technology

DLT has been discussed extensively over the last few years and is considered one of the most hyped contemporary technologies. DLT has been described as a game changer that has the potential to revolutionize many different businesses (Sunyaev, 2020, p. 266). DLT is often understood as a synonym of blockchain, but this understanding is incorrect. The difference lies in the fact that blockchain is a specific type of distributed ledger. Even though the terms “distributed ledger technology” and “blockchain” are often used interchangeably in the literature, it is important to be aware of the slight difference between the two. As mentioned already, blockchain is specific type of ledger in which data can, for example, be stored in a specific format. There are several other types of ledgers, which include different data structures. A ledger (including a blockchain) that is distributed across a network can be defined as a “distributed ledger” or simply as a “ledger” (Chowdhury et al., 2019, p. 167,931). DLT attracted attention after the introduction of blockchain and financial transactions. New DLTs were invented because blockchain was criticized for its cost and its lack of scalability (El Ioini & Pahl, 2018, p. 277). Nevertheless, DLTs that do not rely on blockchain technology are not investigated or discussed further in this thesis. The

thesis only concerns DLT insofar as it pertains to blockchain technology. Therefore, the term “distributed ledger technology,” when used in the chapters that follow, always refers to blockchain technology.

The blockchain, at its core, is simply a record of transaction data. A distributed ledger is a ledger that consists of successive blocks that follow a strict set of rules. Additionally, each block of the ledger, which is distributed and stored in the nodes of a peer-to-peer (P2P) network, is generated at a prespecified interval that is decentralized through the use of a consensus algorithm (Chowdhury et al., 2019, p. 167,931). The consensus algorithm is described in more detail in chapter 2.1.4.

2.1.3 Blockchain: classification

The cryptocurrency Bitcoin relies on a public version of the blockchain. However, blockchains in general can be public or private (O’Leary, 2017, p. 139). In addition to these two types, there are also consortium blockchains. A consortium blockchain is a hybrid of private and public blockchains. To sum up, a distinction is typically drawn between public, private, and consortium blockchains (Shrivastava & Yeboah, 2018, p. 3).

The public blockchain is free to use for any user who wants to participate in the network. In this type of blockchain, there are no restrictions, and any user may participate in the consensus process. Every user can read, submit, and verify transactions (Bernal Bernabe et al., 2019, p. 4). Furthermore, such blockchains are decentralized in their structure. Bitcoin and Ethereum are the most popular examples of public blockchains (Vujicic et al., 2018, p. 5). Public blockchains are considered highly secure due to being decentralized, and the ease of accessing them allows for widespread use. In addition, public blockchains are highly transparent. However, they also have disadvantages, such as high energy consumption and a low number of transactions per second. Private blockchains are centralized or only partially decentralized. Such blockchains can be controlled by a single organization that determines the set of actors who are authorized to read or submit transactions. In addition, that organization also determines who may act as a node and thus acquire authority. Since private blockchains usually have fewer nodes than public ones, validation times are shorter. Furthermore, such a system can be controlled more easily because it is private. However, this also entails disadvantages, in that private blockchains are less trustworthy. In addition, the risk of a security breach is considerably higher (Jo et al., 2020, p. 76). The main difference between public and private blockchains lies in the authorization of individual users. Therefore, classification depends on the set of actors that can access a system and interact with it (Salimitari & Chatterjee, 2019, p. 7).

A consortium blockchain is a hybrid form that combines features of public and private blockchains. The distinguishing feature of consortium blockchains is that the authority to decide on the verification and distribution of the right to read transactions is shared by a group of participants and not assumed by a single central party. However, the blockchain is still not publicly accessible (Jiang & Ding, 2018, p. 2,379). Therefore, consortium blockchains are more decentralized and secure than private blockchains. However, they possess some of the characteristics of private blockchains. Despite higher decentralization, only selected participants hold authorization rights and therefore possess the ability to access such blockchains. In addition, consortium blockchains are more scalable and perform better (Zheng et al., 2017, p. 559).

Private and public blockchains can be differentiated further. Access to transaction validation on a public blockchain can be granted either permissioned or permissionless. All nodes in permissionless blockchains can validate transactions, while only preregistered nodes may do so in permissioned blockchains.

Table 1 below illustrates and overviews the categorizations. The table shows that public permissioned blockchains are open to all. The rights of users include reading the ledger and submitting information to it. However, only authorized nodes can validate transactions and thus write to the main ledger. In contrast, in a public permissionless blockchain, there are no restrictions at all: all nodes can read, write, and validate transactions. A private blockchain, by definition, can only be permissioned. Accordingly, only authorized nodes can read and submit data to the ledger or, of course, validate transactions (Beck et al., 2018, p. 1,022).

Access to transactions	Access to transaction validation	
	Permissioned	Permissionless
Public	All nodes can read and submit transactions. Only authorized nodes can validate transactions.	All nodes can read, submit, and validate transactions.
Private	Only authorized nodes can read, submit, and validate transactions.	Not applicable.

Table 1: Blockchain typology

Source: Beck et al. (2018, p. 1,022)

2.1.4 Blockchain characteristics

The word “consensus,” which appears in the phrase “consensus algorithm,” means “agreement.” One of the main purposes of a consensus algorithm is to maintain the safety and efficiency of a blockchain. Therefore, it is important to choose an appropriate algorithm in order to achieve a significant gain in the performance of a blockchain application. There are two problems that a consensus algorithm must address. The first has to do with the double spending issue: a certain currency may be used in two transactions simultaneously. Blockchain technology resolves this issue by relying on a large number of nodes that are tasked with verifying the transaction. The second problem is called the “Byzantine generals’ problem.” It concerns issues in the distributed system. The nodes that deliver data to one another through peer-to-peer communication can be infected by malicious attacks, which could damage the content of communications. Therefore, nodes must be aware of the problem and be capable of identifying information that has been changed or manipulated. This issue also requires a consensus algorithm to be designed (Mingxiao et al., 2017, p. 2,567). The consensus algorithm, in combination with cryptography, is used to ensure the security of the blockchain. To summarize, the consensus algorithm ensures that all participants receive an exact and identical copy of the blockchain. Different consensus algorithms can be chosen for different types of blockchains. The Proof of Work (PoW) algorithm and the Proof of Stake (PoS) algorithm are the two most used algorithms. Alternatives exist. However, only the most commonly used algorithms are discussed in the following sections (Bach et al., 2018, p. 1,545).

Proof of Work

PoW is a widely employed consensus strategy. It is used in the blockchain network of the cryptocurrency Bitcoin (Zheng et al., 2017, p. 599). The main idea is to allocate accounting rights and rewards to nodes. Those rights and rewards are contested through a hashing power competition. The nodes calculate a

difficult mathematical problem, which is based on a previous block. For example, in the Bitcoin blockchain, the first node that solves a problem can create a new block and therefore acquire a certain amount of bitcoin as a reward (Mingxiao et al., 2017, p. 2,568). Specifically, the hash value of the block header is calculated by each node in the network. In order to achieve different hash values, the miners change the nonce, which is contained in the block header, frequently. The calculated value should be equal to or lower than a given specific value in order to comply with the consensus algorithm. To confirm the correctness of the hash value, a node must reach the target value to transmit the block to other nodes. Thereafter, once the block has been validated, other miners will add the newly created block to their own blockchains. The nodes that participate in hashing power competitions and calculate hash values are called “miners.” When multiple nodes almost simultaneously obtain the suitable nonce, valid blocks are generated simultaneously in a decentralized network. However, the probability of two competing forks generating the next block at the same time is small. Furthermore, in the PoW protocol, the chain that becomes larger or longer is subsequently seen as the authentic one (Zheng et al., 2017, p. 560). A disadvantage of the PoW method is that solving trivial puzzles requires enormous raw computing power and leads to massive energy consumption due to the competition between validators (Saleh, 2021, p. 2). The larger the number of individuals who participate in the competition, the more computing power is required to solve the problem, which exacerbates the problem. Due to the need for more computing power and the formation of professional mining pools, mining bitcoin has become unattractive for private individuals and is controlled by large organizations. For a PoW blockchain, it is important that none of the participants own more than 50% of the computing power of the whole network. Otherwise, double spending attacks would become possible, and it would be impossible to guarantee security from manipulation (Shi, 2016, p. 1).

Proof of Stake

In the last few years, PoS protocols have been investigated actively. The question is whether PoS can replace the largely inefficient PoW mechanism, which is currently used in most open blockchains. PoS can be seen as a strong alternative to PoW, and it can remedy the issues that plague PoW. One of the main problems of PoW is that it involves an extravagant waste of energy as a result of electricity consumption on the part of miners (Li et al., 2017, p. 297). In contrast, PoS networks remove the need for solution searching, and block leaders are no longer chosen by computational power. In the new PoS process, block leaders are selected on basis of their stakes. Since the stake-based leader selection process does not depend on computational power, considerably less energy is wasted (Nguyen et al., 2019, pp. 85730-85731). Unlike the PoW protocols, PoS protocols do not cast active nodes as miners. Instead, they are called “validators” or “stakeholders” (Maung Maung Thin et al., 2018, p. 198). Furthermore, PoS is also seen as being much faster than PoW. For the PoW mechanism to remain secure, the speed of block generation and transaction confirmation must be kept low because the miners propose many different blocks. In contrast, under the PoS mechanism, only one block is created in each round. Therefore, transaction-confirmation and block-generation speeds are much higher. Accordingly, the PoS mechanism has gained popularity. As mentioned previously, PoS protocols are more energy efficient, which was also a reason for developing an alternative to PoW. In this alternative algorithm, the number of digital tokens is called the stake of a node. The digital tokens can, for instance, be coins in cryptocurrency that are held or deposited. The searching process consumes little energy. The searching process in PoS is also different, in that a leader is chosen, based on stakes, to execute the mining process and to create a new block. The stake of a node influences the chance of being selected as the next block leader. The higher a stake, the higher the chance of being selected as leader (Nguyen et al., 2019, p. 85,731). In other words, the users or nodes that validate transactions are selected on the basis of their wealth, which

is defined as a stake in this context. Additionally, the validators are rewarded with a share in each transaction that they validate (transactions fees; Deirmentzoglou et al., 2019, p. 28,712). However, PoS protocols also have downsides. These disadvantages are mainly related to security threats. Nothing-at-stake and long-range attacks are two well-known threats. The former makes it possible for nodes to mine conflicting blocks without risking their stakes, which leads to an increase in the number of forks in the system. Furthermore, reaching consensus takes more time. The second attack, which is also known as a “history attack,” involves an adversary that seeks to modify the entire history of the blockchain. This modification starts with early blocks and can even include the genesis block. Such scenarios become possible when the adversary obtains the private keys of past validators (Li et al., 2017, p. 298).

2.1.5 Smart contracts

The term “smart contracts” was used at the end of the 1990s by the American jurist Nick Szabo. The idea was that contracts could be represented in terms of software and hardware so that performance and consideration would be guaranteed by a predetermined program logic. The idea is that dispensing with human authority in the execution of a contract would reduce risks and transaction costs. Furthermore, the contracting parties do not have to trust each other. They merely have to believe in and rely on the machine, which performs the rest of the work (Kaulartz & Heckmann, 2016, p.618).

One example of a smart contract would be the leasing of a car whereby the car can only be used for as long as leasing instalments are paid. In this scenario, a smart contract that is executed on the car’s board computer can monitor compliance with the programmed condition (“Have all due leasing instalments been paid?”) via an interface with the bank or directly via blockchain. The car’s ignition can only be activated when the condition is met. Additionally, the car, if the user charges it with money through transactions, can also pay for parking space independently. The car and the parking machine would communicate with each other automatically about the conditions of parking. The foregoing can be seen as a prime example of machine-to-machine (M2M) communication. M2M communication entails automated communication between two machines without the intervention or participation of a human being. It entails the free, direct, and automatic transmission of information or data between two machines or electronic devices. For a long time, there were no use cases for smart contracts. The available technologies did not seem suitable for achieving the two main purposes of smart contracts, namely increased contractual security (relative to traditional contract law) and the reduction of transaction costs. However, this state of affairs changed with the emergence of blockchain technology. Users increasingly realized that the promise of blockchain extends far beyond cryptocurrencies such as Bitcoin or Ethereum. Blockchain also enabled the possibility of exchanging and managing digital goods through smart contracts by creating the ideal conditions for this functionality to be implemented (Kaulartz & Heckmann, 2016, p. 618).

In her book, Finck (2019) defined a smart contract as a program code that runs on a blockchain and reallocates digital assets or representations of physical objects. In other words, the author described the rights to those assets passing between two or more parties in the form of transactions on the basis of external data that are not known at the time when the code is written (Finck 2019, p. 4). Therefore, smart contracts are codes that are executable and run on a blockchain to facilitate, execute, and enforce agreements between actors that do not trust each other without the need for third-party authority (Khan et al., 2021, p. 2,902).

Furthermore, Finck (2019) explained that the lifecycle of a smart contract that is based on blockchain technology has four stages, namely “creation,” “freezing,” “execution,” and “termination.” During the creation phase, the smart contract is defined and converted into code. In the next stage, the software is frozen while it is added to the chain through the relevant consensus process. This occurs before the software is executed, that is, prior to its reading and implementation by the various nodes. The process ends when the new-state information and transactions are stored in the blockchain and when they are confirmed according to the consensus protocol (Finck 2019, p. 4).

Wang and Nixon (2021), complementing this idea, described smart contracts principally as computer programs that verify or implement a contract that is capable of executing automatic predefined conditions without the involvement of a third party. The result (or output) of a smart contract can be treated as a token. The two authors used Ethereum as an example. Ethereum provides a platform that can be used to create arbitrary smart contracts. These Ethereum tokens, as a smart-contract output, can be utilized to represent different digital assets as well as physical and virtual objects. Furthermore, the tokens have highly diverse purposes such as recording transactional data or paying for access to a network. Initially, its development is unprogressive to establish a mapping process from the representative asset to its token form. The distinctive property of tokens that emerges from the model of the representative asset is validated by the concept of smart contracts. Tokens do not have any physical content and do not depend on an operating system, in the general sense of that term. However, the smart contract enables the validity of a token to be verified easily (Wang & Nixon, 2021, p. 3).

2.1.6 Understanding of a token

A token is a digital asset that is stored on a blockchain or as cryptocurrency. The notion of a token, however, is not unique to the blockchain. Tokens have historically been used to safeguard digital exchanges such as banking transactions. At present, tokens are considered to be one of the most valuable and important applications of blockchain technology. It is for this reason that tokens have been described as “central to most social and economic innovations developed with blockchain technology” (Wang & Nixon, 2021, p. 3). Specifically, tokens can represent various assets or rights that are digital, physical, virtual, or legal. In addition, they can usually be traded easily and rapidly, with low transaction costs (Treiblmaier, 2021, p. 2). Crypto tokens are often a programmable asset that is managed by a smart contract, which is used within a project or dApp. The term “dApp” describes an application in a P2P network that is not controlled by a single entity and is therefore referred to as a decentralized application. Tokens can serve as a means of payment, that is, as currency. In such cases, they are referred to as the “local currency” of a dApp. However, crypto tokens have many properties that do not pertain to exchange functionalities. Their outstanding programmability is one of those properties. In this context, they are used in order to trigger certain functions of the smart contracts in the dApp. In addition, tokens can be linked to off-chain assets. They can be used as means of fundraising, of building an ecosystem or a community, and of pre-ordering or investing. The tokens can be created on an existing blockchain through smart contracts, which are called token contracts. This is a widely used and accepted application. Therefore, coding patterns and best practices are available. There are also token contract manufacturers, which can be found on-chain or as web services. There are different ways of acquiring tokens. For instance, they can be purchased in the course of an initial coin offering (ICO), which is discussed in more detail in Section 2.1.7. They can also be bought on a crypto exchange, traded on the chain, or received for free in the form of an airdrop as a reward for a specific activity or service. The value of a token is primarily determined by supply, demand, and community belief, which reflects its credibility and usefulness (di Angelo & Salzer, 2020, p. 1).

There are several types and categories of tokens. Accordingly, Freni, Ferro, and Monacada (2020) attempted to classify tokenization. They wrote that acquiring an in-depth understanding of the nature of a token and the ability to describe it comprehensively is as crucial as it is challenging. They believed that two questions must be considered when classifying a token. What is the function of the token? What does the token represent (Freni et al., 2020, p. 2)? A widely used categorization distinguishes between payment tokens, security tokens, and utility tokens, which are classified in greater detail below. Drawing a distinction is necessary because security tokens are more heavily regulated than other tokens in most jurisdictions (di Angelo & Salzer, 2020, p. 2).

Three types of tokens can be identified.

1. Payment token

Payment tokens are known as payment or cryptocurrency tokens. They are mainly used for transactions between users. These tokens are designed for trading and expenditure (FINMA, 2018, p.2; BaFin, 2019, p. 6).

2. Utility token

Utility tokens are virtual vouchers that are supposed to provide holders with functional benefits in the form of access to a network. These tokens can carry special rights, such as the ability to exchange tokens for products or services at a specific time (FINMA, 2018, p.2; BaFin, 2019, p. 5).

3. Security token

Security tokens represent specific investments, such as shares or voting rights in a company or another organization and tangible or digital stores of value. Security tokens not only serve as a digital representation of an underlying asset or benefit but can also be programmed with a vast array of unique features and property rights. These tokens are an entirely new type of digital asset (FINMA, 2018, p.2; BaFin, 2019, p. 6).

There is no international or generally accepted classification of tokens yet. Therefore, the token categorization above is intended primarily as a simplification that reflects the literature and covers the main tokens that are currently in circulation. In practice, each token, regardless of its form or designation, must always be examined and evaluated by reference to its content. Hybrid tokens may also exist (Markheim & Berentsen, 2021, p. 63). Wang and Nixon (2021) confirmed that, from an application perspective, there is no internationally recognized categorization of tokens. However, the literature does provide some classifications that are based on functionality. Furthermore, it describes a variety of types of tokens that have different applications in the blockchain tokenization process. Instead of discussing these specific application-based tokens, authors typically focus on another classification, which is based on the fungibility of assets in blockchains. The fungibility classification is, in rough terms, threefold, and it refers to fungible tokens (FTs), NFTs, and semi-fungible tokens (SFTs; Wang & Nixon, 2021, p. 3-4). It is thus possible to classify tokens in different ways. This additional classification is important for this thesis because it discusses tokenization in the context of virtual real estate. NFT hype occupies a central place in the discussion.

The fungibility classification has three categories.

1. Fungible tokens

An FT can be replaced by another asset of the same category (Wang & Nixon, 2021, p. 4). FTs can thus be exchanged for an asset of the same value because they are not unique and can be divided or merged (Regner et al., 2019, p. 2). Therefore, FTs can be defined as interchangeable

(Kiong, 2021, p. 21). For example, Wang and Nixon (2021) stated that an FT can be replaced by another type of asset of equal value and that it can be divided into smaller parts without its original value changing. The authors also referred to these types of tokens as identical to others and wrote that dividing them into smaller units does not affect their value. In another example, they addressed non-existent uniqueness. In their example, they described payment tokens as fungible because they are exchangeable, divisible, and not unique in essence. FTs have a limitation: many valuable assets, such as art or real estate, cannot be replaced or divided. Therefore, these valuable assets must be represented by NFTs (Wang & Nixon, 2021, p. 4).

2. Non-fungible tokens

Unlike FTs, NFTs are cryptographically unique and cannot be replicated. These features make it possible to keep track of the ownership of individual assets (Wang & Nixon, 2021, p. 4). Therefore, NFTs entail the existence of a specific authority that has exclusive ownership rights over the digital asset. For example, a random internet user can view an NFT online, but only the individual who bought that NFT is its owner (Kugler, 2021, p. 19). Thus, NFTs differ from FTs in their interchangeability, divisibility, and uniformity. It is not possible to divide an NFT into smaller parts because each token contains unique information and attributes that distinguish it from others. Therefore, NFTs are not interchangeable. In summary, every NFT is unique and distinct (Wang & Nixon, 2021, p. 4). Regner, Schweizer, and Urbach (2019) wrote that NFTs can represent both physical and virtual goods. They cite luxury goods in general and automobiles in particular as examples of physical goods that can be represented by NFTs. Writing on virtual goods that may be combined with NFTs, they referred to virtual gaming assets, digital artwork, and software licenses (Regner et al., 2019, p. 2)

3. Semi-fungible tokens

SFTs combine features of FTs and NFTs. They are a new class of token. SFTs provide additional flexibility in the representation of complicated assets or processes. The fact that SFTs contain and represent the characteristics of both fungible and non-fungible assets makes it possible for them to be used more efficiently when a token transaction is about to be created and bundled. Therefore, it is not necessary to create an individual contract for each token, which makes operations much more convenient (Wang & Nixon, 2021, p. 5). Singh and Singh (2021) wrote that SFTs are sets of tokens that serve the same purpose or goal. Writing on use cases, they mentioned items that are owned by a single entity, form part of a series, and have a serial number or other metadata that differ only slightly, such as concert tickets and 3D games (Singh & Singh, 2021, p. 317). Bamakan et al. (2022) further explained that SFTs can be convertible within a class, have the same intent or purpose, and exist for a specific time. They mentioned airline tickets as a simple example. Airline tickets can be considered as SFTs. An economy ticket can be easily exchanged for another economy ticket but not replaced with a first-class one. SFTs have blockchain-based games as their main application and decrease the overheads that are associated with NFTs (Bamakan et al., 2022, p. 3)

It is also important to draw a distinction between coins and tokens. The two terms are often used synonymously because they exhibit similarities. However, close scrutiny reveals significant differences. Tokens do not have their own blockchains and are therefore constructed on top of existing ones. This is exactly what differentiates them from coins (di Angelo & Salzer, 2020, p. 1). Therefore, at the highest level, there is a distinction between a cryptocurrency that has its own blockchain platform (a coin) and one that runs on an existing blockchain (a token; Amsden & Schweizer, 2018, p. 10).

2.1.7 Initial coin offerings and security token offerings

The term “initial public offering” (IPO) describes the process of bringing the shares of a private company into circulation. In an IPO, those shares are placed on a global stock exchange for the first time. This process occurs primarily as an attempt to raise capital (Jain & Kini, 1999, pp. 1281-1282). An ICO is akin to an IPO (Arnold et al., 2019, p. 235). Depending on the type of token that is being issued, it can also be referred to as a “security token offering” (STO; Guides, 2019).

In the literature, ICOs are often compared to crowdfunding. Amsden and Schweizer (2018) described ICOs as a modern and unregulated approach to crowdfunding that revolves around exchanging risky tokens for legal tender or cryptocurrencies through a blockchain (Amsden & Schweizer, 2018, p. 1). Writing on this matter, Arnold et al. (2019) investigated and compared the concepts of ICOs and crowdfunding. They found several similarities and differences. In terms of similarities, both are intended to raise venture capital in order to enable the company to continue to grow or to finance new projects. Both include a public fundraising call that gives practically everyone the opportunity to participate in the investing process. In addition, as the internet is a primary method of payment and communication, anybody can contribute from practically anywhere in the world. Finally, the authors mentioned that contributors typically obtain something in return for their investment. The consideration can take the form of hardware, a token with a specific function, or equity in the company. Turning to dissimilarities, ICOs and crowdfunding rely on different technologies. While ICOs make use of decentralized peer-to-peer networks and blockchain technology, traditional crowdfunding employs centralized platforms managed by third party providers. Furthermore, in blockchains, the completion of financial transactions does not depend on the existence of a trusted environment. Therefore, no trust in any entity is required. In crowdfunding campaigns, conversely, platforms and banks offer trustworthy environments. In ICOs, transactions are simply verified by a network-wide consensus mechanism (Arnold et al., 2019, pp. 235-236).

It might be desirable to examine ICOs in greater depth. In the case of an ICO, investors purchase tokens that are issued by a start-up firm through a platform. This transaction takes the form of a token sale, which is offered by the start-up firm in return for cryptocurrencies (most commonly Ethereum or Bitcoin) but also, rarely, for fiat money (e.g., euros). From a technological standpoint, cryptocurrency moves from an individual's wallet to a smart contract that is made by the issuing corporation (Hahn & Wons, 2018, p. 1). Initial Exchange Offerings (IEOs), an alternative to ICOs, emerged in 2018. In an IEO, tokens are issued directly via the trading platform, on which they are subsequently listed and traded. The direct involvement of the cryptocurrency exchange makes it the most important marketing partner of a project. In addition, the direct contribution of the exchange to the choice of projects, the organization of the token sale, and the sale of tokens makes these investments more attractive (Myalo & Glukhov, 2019, p. 13). In general, trading tokens for crypto coins has become a practical and popular alternative form of financing for start-up companies. One obvious reason that might prompt a company to develop a token that represents the rights and obligations of token holders is that it enables the firm to avoid the cumbersome and strictly regulated path of traditional venture capital financing, which entails multiple rounds of equity investment by venture capitalists. The Ethereum platform provides a protocol for creating such tokens (the ERC token standard) that requires less than 100 lines of code. Token creators write white papers to outline their business ideas (Hacker & Thomale, 2017, p. 11). A white paper has two main objectives: it provides information to potential investors, and it functions as an advertisement for the token. Although writing a white paper is a matter of choice, white papers are often considered necessary or even mandatory from the viewpoint of the ICO issuer (Kasatkin, 2022, p.

2). Thus, the quality of the white paper that a group of developers publishes has a significant influence on the success of the token sale, a point that has been confirmed by a recent study (Hacker & Thomale, 2017, p. 43). Beyond the aforementioned business plan, the white paper also contains information about the design of the blockchain platform, the number of units to be sold, their price, a description of the developer team, an outline of milestones, and such like (Kasatkin, 2022, p. 2).

2.1.8 Regulation

Those who regulate the blockchain world, where the focus is primarily on tokenization and tokens, must overcome many hurdles. Tokenization is considered to possess the potential to revolutionize the financial sector and to improve investment processes by reducing complexity and providing more opportunities for investment. For this potential to be harnessed, existing regulatory ambiguities must be addressed (Sunyaev et al., 2021, p. 473). According to research, blockchain finance must prioritize more effective privacy and security mechanisms. In addition, there is a need for improved financial transaction regulations that not only favor ICOs but also protect investors (Gorkhali & Chowdhury, 2022, p. 47).

At present, there is no uniform legal basis on which the member states of the EU can eliminate or mitigate these regulatory uncertainties. However, the European Commission has published a draft regulation with new rules for the issuance and trading of tokens, the MiCA (Markets in Crypto-Asset regulation) proposal. The proposal aims to govern the distribution, trading, and issue of digital currencies and crypto assets in the EU. Its objective is to create uniform regulation for digital assets and thus a trading space. Furthermore, given the rapid developments of the crypto market, a basic framework for its further development within the EU should also emerge. A cross-border legal regulatory framework with an adequate and audit-compliant infrastructure could be a game changer in the crypto field (Bočánek, 2021, p. 38). Legal definitions and specific standards should establish a functioning set of rules and a regulated market. A classification of crypto assets is proposed. Each category carries distinct legal implications. The proposal still has several shortcomings, and there is room for improvement. Additional specifications are needed, and some gaps must be covered. However, the quality of the regulation is considered to be high, especially since it is the first proposal to legislate in a wholly unregulated domain (Bočánek, 2021, p. 50). One reason for the high quality of the proposal could be the blockchain law (TVTG) that came into force in Liechtenstein. The European Commission treated this law as a model, and it adopted some of its elements, such as the token container model, the requirement for authorization as a condition for the provision of certain blockchain services, and the information requirements for public offerings (Niedermüller, 2021). This section discusses the Liechtenstein blockchain law in detail.

Liechtenstein was one of the first countries to address the regulation of the token economy through a legal framework and thus became a pioneer of the legislative effort (Teichmann & Falker, 2021b, p. 784). The Blockchain Act uses the term “trustworthy technology” (TT) and therefore applies to all TT service providers. This abstract term was chosen instead of “blockchain” or “distributed ledger technology.” The Act came into force on 1 January 2020, and token issuers, token generators, TT key depositaries, and TT token depositaries had to register with the Financial Market Authority of Liechtenstein (FMA) on a professional basis (Shashkova et al., 2020, p. 30). The Liechtenstein government provided a more general legal framework through the Blockchain Act, which aims to cover all possible applications of blockchain instead of focusing exclusively on cryptocurrencies. Given the rapid emergence and constant evolution of technology, the Tokens and TT Service Providers Act (German: *Token- und VT-Dienstleister-Gesetz*; TVTG) is intended to provide a comprehensive legal framework that can keep

pace with innovation. Therefore, the abstract term “trustworthy technology” was used to ensure that the law would have a broad scope of application.

The new law also amended existing statutes to promote legal certainty in a token economy. It thus addresses the due diligence obligations from the Due Diligence Act (German: (*SPG*) *Sorgfaltspflichtgesetz*) of 2008 as they relate to cryptocurrencies. Since September 1, 2017, the SPG have controlled businesses that exchange regular currencies for cryptocurrencies, and all transactions of a value in excess of 1,000 francs are subject to the due diligence laws (Art. 3 SPG). The TVTG also established a registration system for TT service providers (Teichmann & Falker, 2021b, p. 784).

Since the TVTG is not part of Liechtenstein financial market law, TT service providers are not considered financial intermediaries and are not subject to the same licensing, registration, and supervisory requirements as financial intermediaries. The due diligence provisions are an exception, in that they apply to both financial intermediaries and TT service providers. The TVTG imposes some basic conditions on all applications of blockchain that are not covered by the due diligence requirement or the FMA regulations. These criteria apply to all TT providers and are designed to safeguard consumers as well as Liechtenstein's global reputation by ensuring compliance with international standards (Teichmann & Falker, 2021a, p. 98).

The main attraction of the TVTG is the introduction of the token container model (TCM). Tokens are defined in an innovative way, and no categorization of tokens is required. In this model, a token can represent any right. Therefore, traditional utility, security, or payment tokens are all captured by the TCM. This feature makes the model unique: it encompasses all forms of property, from commodities and securities to intellectual property. The model captures all kinds of rights, which reflects the ever-changing world of blockchain technology. As a result, the TCM protects both existing rights and rights that are related to blockchain technology (Nägele, 2020, p. 565). In addition, the TVTG introduces the role of a “physical validator,” which is intended to act as a bridge between the offline world and the internet. The task of the physical validator is to verify that the rights to the property that a token represents are contractually enforced within the meaning of property law. In this context, the TVTG also establishes a civil-law foundation for ascertaining the underlying right to the token and for confirming its effective transfer from one party to another. It also defines “effective transfer” in the context of new digital assets. Users who lack experience in storing or transferring tokens can delegate these tasks to registered service providers (Nägele, 2020, p. 566). The tokenization of shares supplies a salient example. A physical validator must ensure that the share exists in the physical world. The physical validator is therefore a living individual who ensures that only existing objects and rights are transmitted into the digital domain. In addition, the physical validator must take custody of the physical certificates to prevent digital and conventional assets from being sold separately (Layr, 2021, p. 65).

Switzerland is considered a pioneer in the crypto world and is thus engaged in advancing the innovative blockchain technology, including its potential. For this reason, the Swiss authorities are trying to offer the best possible regulatory framework and, eventually, to inspire cross-border laws (Frick, 2019, p. 99). The main focus of the FINMA (German: *Eidgenössische Finanzmarktaufsicht*, English: *Swiss Financial Market Supervisory Authority*) I guidelines is on anti-money laundering and securities regulation. These areas of law are most relevant to ICOs (Thevenin & Schrag, 2018, p. 55). Therefore, FINMA, the Swiss financial sector regulator, published guidelines on ICOs on 16 February 2018. In these guidelines, FINMA distinguished between three different categories of ICOs. The first category consists of “payment ICOs,” which simply serve the function of payment. Next, “utility ICOs” are about digital access

rights in applications or services. The FINMA also referred to “asset ICOs,” which give rights to earnings (Kondova & Simonella, 2020, p. 45). Projects that would be covered by the Banking Act, which regulates deposit taking, or the Collective Schemes Investment Act, which regulates investment fund products, are now regarded as uncommon. The Anti-Money Laundering Act is designed to safeguard the financial system against the risk of money laundering and funding terrorism. These risks increase in a decentralized blockchain-based economy because assets may be moved anonymously and without the use of regulated middlemen. At the same time, security regulation ensures the transparency and comparability of financial statements, an adequate minimum standard of disclosure, and fair and efficient price formation through secure trade (Thevenin & Schrag, 2018, p. 55). FINMA have also resolved to classify tokens according to their functions, which should result in differential treatment. When categorizing tokens, it is important to understand their purpose and design. Thus, the categorization focuses on the purpose of the tokens and their tradability or transferability. Once again, FINMA distinguished between three types of tokens while admitting to the possibility of hybrid forms. The token categorization is at the core of the ICO classifications that were mentioned above, and there is an obvious connection between the two. Tokens that are categorized as payment tokens are transferable and can be used to pay for products or services now or in the future as well as to store or transmit quasi-monetary value. This category includes any cryptocurrency that does not give rise to claims against its issuer, such as central bank-issued notes and ledger accounts. According to the FINMA guidelines, payment tokens are not treated like financial securities. However, transactions must comply with money laundering regulations. Utility tokens provide users with digital access to an application or a service that runs on a blockchain. However, they may be considered as securities unless their sole purpose is to grant digital access rights. The FINMA Transaction Purpose Matrix indicates that they may be considered securities if they are used partly or exclusively as investments. Finally, assets such as a debt or equity claims on issuers are represented as asset tokens (Thevenin & Schrag, 2018, pp. 54-55). Such claims may result, for example, from promises of shares in future company profits or future capital gains. Assets tokens exhibit significant functional analogies to securities

if they pay dividends or interest or represent rights to earnings streams, with or without claim to repayment of principal, or confer voting rights, or if they are in their economic function analogous to equities, bonds, or derivatives, or other structures capable of securitization that enables physical assets to be traded on the blockchain (Thevenin & Schrag, 2018, p. 54).

Therefore, asset tokens must adhere to all securities rules and regulations. There are also hybrid tokens, which combine at least two of the characteristics from the three categories that were mentioned above. All rules that apply to at least one of its components must be fulfilled. A utility token, for instance, has payment-token features and is thus governed by anti-money laundering legislation. The sudden increase of hybrids in creative financial and legal engineering is both obvious and predictable. FINMA is therefore trying to promote the potential of hybrid token business models and the resulting ICOs without stifling them. Overzealous regulation has already stifled innovation in other countries, such as the UK, South Korea, and, partly, the USA. Companies have thus been driven to countries like Switzerland that promote innovation through their regulation (Thevenin & Schrag, 2018, p. 54).

Switzerland continues to advertise itself as a leading blockchain jurisdiction. On August 1, 2021, Switzerland became one of the first countries to enact a legal framework for blockchain technology. Through that enactment, Switzerland improved the certainty of its laws and enabled further growth and innovation. The process focused on modifying existing laws rather than on the passage of new legislation. The core of the bill focused on amendments to securities law. The law on the infrastructure of the financial market and bankruptcy law were also adapted. These developments provided a strong legal foundation

for the trading of rights on electronic registers. A new license category for DLT trading systems was also introduced. It creates a legal framework that can be adapted to new types of financial market infrastructure (SIF, 2022).

2.1.9 Tokenization

In essence, tokenization is a process whereby an issuer creates multiple digital tokens on a blockchain (Popescu, 2021, p. 26). Therefore, tokenization is primarily about tokens, which are usually digital representations of an asset that exists in the physical or the virtual world. In the blockchain space, for example, a token could be used to represent cryptocurrencies such as Bitcoin or Ethereum. From a technical perspective, tokens are created by an algorithm that is defined in a smart contract (as discussed in the previous chapter), which runs on a blockchain. Thus, the process by which data or assets are transformed into tokens, which are basically random digitized sequences of characters, is called tokenization (Wang & Nixon, 2021, p. 3). The underlying idea is to eliminate the need for a trusted third party that authenticates a transfer of ownership between unknown parties. Traditional operations are frequently associated with high costs, lengthy processes, and single points of failure. These issues are resolved through a decentralized solution that takes the form of tokenization (Heines et al., 2021, p. 3). The process of representing physical or virtual assets is simplified by tokenization. In addition, it provides some protection for sensitive data, for example by substituting non-sensitive data. The token that is created cannot be used to determine the values of the original data or that of the assets in blockchain applications; it only serves as a reference. Therefore, the token holds no information of economic value, and its price is determined by the market. Consequently, a token can be conceptualized as a symbol that is validated by the smart contracts of the target blockchain system. The token can be utilized in a variety of applications or exchanged on the market. It is authenticated by the smart contract. The trend toward tokenizing real-world assets has sparked considerable interest in blockchain research (Wang & Nixon, 2021, p. 3).

The benefits of tokenization on the blockchain are numerous. However, the blockchain tokenization process is still in its infancy and faces many challenges, including regulatory and technical ones, in order to fully achieve the potential of tokenization in combination with blockchain (Wang & Nixon, 2021, p. 3). As far as the technological challenges are concerned, multiple studies have shown that technical immaturity causes applications to become stuck in the prototyping stage. Other publications have claimed regulatory, operational, and governance hurdles must be overcome in order to ensure the widespread adoption and commercial acceptance of tokenization. In particular, the legal concerns and transition risks that are linked to the issuance of digital securities must be considered adequately within national legal systems. Current research on the application of tokenization focuses mostly on general opportunities and challenges, with a strong focus on the financial sector (Heines et al., 2021, p. 4).

The technical components of tokenization are described in detail in the following paragraphs. The exposition focuses on the Ethereum platform because of its use of the ERC20 token standard, which is one of the most popular types of Ethereum smart contract. This token type alone attracts more than 100 billion dollars of transaction volume daily. Thus, the Ethereum platform is considered a popular medium for tokenization. It also offers standards other than the ERC20 (Ivanov et al., 2021, p. 22). Since tokenization on the Ethereum platform is popular and used widely, the technical process that is associated with the Ethereum platform is described here. The token standards on the Ethereum platform provide basic functions and are therefore smart contracts. The most crucial purpose of a smart contract for tokenization

is to keep track of ownership in the tokens that are created. The public Ethereum address that is associated with a token can be modified by using a transaction message. A compatible wallet is required for a user to hold and transfer a token. A wallet is a software application that enables tokens to be received and dispatched easily. Specifically, it is used to store private keys (Markheim & Berentsen, 2021, p. 70-71).

Transactions can be signed and authenticated using a private key, which acts as a unique credential for the blockchain asset. As a result, if the private key is lost, so is control and ownership of the account. Thus, private key leaks are one of the most significant issues for blockchain security (N. Wang et al., 2019, p. 542). The importance of private keys makes them lucrative targets for attacks—possession of a private key means that an attacker effectively controls all of the currencies in the exposed wallet. In order to prevent such a scenario from occurring, it is recommended that wallets be maintained offline on isolated air-gapped computers. The term “cold wallets” is often used to refer to air-gapped wallets (Guri, 2018, p. 1,308).

The sender uses the wallet to send a transaction message in order to transfer ownership of a token. Among other things, the message also includes the address of the receiver, the contract address, and the number of tokens to be transferred. After only several seconds of the transaction message being sent, the transfer of ownership is usually registered in the Ethereum blockchain. The token issuer must take various parameters into account when designing a token, of which the number of tokens that are created, the number of tokens that are offered and available for sale, the issue price of the tokens, their divisibility, the permissible methods of payment, the sales window, and possible trading limitations and restrictions are among the most essential. If the tokenization is conducted through the Ethereum blockchain, ether is usually accepted as a form of payment. However, it is common practice for the issuer to also permit payment in fiat currency. Nevertheless, the buyer must provide an Ethereum address to which the acquired tokens can be sent regardless of whether payment is made in ether or in a fiat currency. Furthermore, the issuer can limit the tradability of a token through a smart contract, which does not require much effort—all that is needed is a whitelist. A whitelist lists all public addresses that have the authority to receive tokens. The smart contract that manages token ownership thus only enables transactions between addresses on the whitelist. Tradability restrictions often result from regulatory requirements, which vary between countries (Markheim & Berentsen, 2021, p. 71).

2.2 Tokenization of physical real estate

Physical real estate is considered illiquid and lumpy. For the tokenization of real estate assets to develop, there needs to be demand for the fractionalization of individual properties. Awareness needs to be raised among market participants, which requires a basic understanding of blockchain technology (Baum, 2021, p. 41). The goal of tokenization, when it is applied to physical real estate, is the fragmentation of an illiquid asset into exchangeable and tradable virtual tokens. As already mentioned in the section on tokenization, the Ethereum blockchain has established itself as the standard for issuing tokens. Therefore, the ownership rights to these tokens are recorded on the Ethereum blockchain and can be traded on decentralized trading platforms. Thus, these tokens are created in accordance with the ERC20 standard and are also referred to as “security tokens.” The maintenance of ownership rights is the key difference between a traditional security and a security token—the latter can only be transferred by its owner (Markheim & Berentsen, 2021, p. 68).

When tokenizing a physical real estate, the asset becomes a digital representation of the asset, with each token representing a portion of the property. It is also referred to as converting the rights of an asset, such as a physical property, into a process facilitated and enabled by the blockchain platform. Precisely because it is very difficult to divide physical assets such as real estate, digital tokens enable consumers and sellers to acquire direct ownership. These digital tokens can be traded like traditional securities on a secondary market. Figure 2 illustrates the principle of tokenization of physical real estate in a simplified way. A property valued at \$50 million is used as an example in this figure. Furthermore, it was determined that the property should be represented by 1 million tokens, which should be issued. Based on this scenario, each token is valued at 50 dollars and can be traded on a secondary market. However, this is only one way of tokenizing physical real estate. Besides direct tokenization, there are also other ways to tokenize real estate, in which the digital tokens represent dividends, distributions, profit participation and property rights and so on (Tian et al., 2019, p. 3).

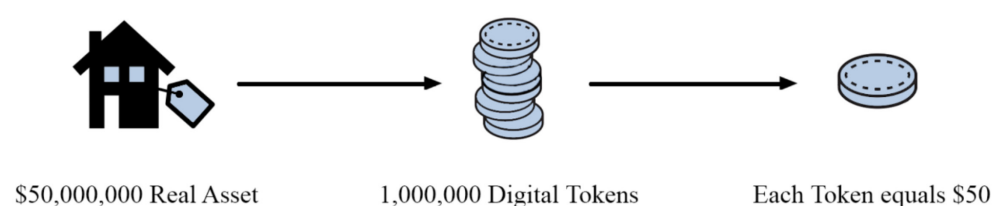


Figure 2: Principle of tokenization

Source: Tian et al., (2019, p. 3)

How does a physical object, such as real estate, become tokenized in the first place? Usually, four stages must be completed in the process of tokenization, namely asset identification, asset appraisal, tokenomics, and smart contract creation and audit. Asset tokenization starts with asset identification. The asset is defined in detail in this step. In the evaluation step, most assets must be appraised by a reputable third party. In general, the valuation of the token is included in the whitepaper for its sale. The third step is the most challenging. All token parameters must be defined, and answers must be given to questions such as “what happens to unsold tokens?” and so on. The final phase includes smart contract creation and auditing. However, before a smart contract can generate tokens, a blockchain must be chosen for the operation (Sazandrishvili, 2020, p. 71). As noted previously, the Ethereum blockchain is the industry standard and therefore the most likely choice (Ivanov et al., 2021, p. 22). Because the conditions of a smart contract cannot be amended once it is live on the blockchain, it is advisable to engage an expert early in the smart contract construction process to avoid complications (Sazandrishvili, 2020, p. 71). Tian et al. (2020) wrote that the verification of requirements and regulatory guidelines as well as the observance of compliance requirements are also important steps in the tokenization process (Tian et al., 2020, p. 6).

2.2.1 Real-world examples of applications

In 2018, the Aspen coin launched one of the first real estate STOs successfully. The provider intended to create a unique real estate offering on the market. They enabled individual investors to own shares in a highly attractive asset that had previously only been available to institutions. Aspen redefined this model by providing individual investors with liquidity options for a single-asset investment (Soltanifar et al., 2021, p. 225). Meanwhile, the number of providers worldwide has increased. There were roughly 41 providers of tokenized real estate at the end of 2021. Europe was home to more than half of them. Small investors can use tokens to invest in residential and office buildings through these companies (Melichar, 2021). Therefore, this section provides real-life examples of the first property tokenization

operations. The few companies that operate in this sector and several real-life cases are presented in order to demonstrate the practical implementation of tokenization and to overview the current market.

Crowdlitoken

Crowdlitoken AG was formed in 2018 to allow small investors to participate in real estate digitally, with prices starting at CHF 100.-. Crowdlitoken AG is one of the first European companies to offer the possibility of investing in a real estate portfolio via a crowdfunding platform by means of an STO. The business model therefore provides small investors with access to co-ownership shares in high-yielding investment properties in Switzerland for little consideration. The users of this platform have the opportunity to purchase the CRT after the registration process has been completed. The CRT is the issued token, which is a legally approved digital bond. The principal features of this bond are such that it combines the advantages of direct and indirect investment. Crowdlitoken's STO can be conceptualized as Crowdlitoken AG issuing tokens as digital representations of their obligation. In this sense, the STO can be seen as an issuing process through which funds are raised and then invested in real estate in accordance with a portfolio strategy. In this context, it must be noted that Crowdlitoken AG remains the owner of the property. Thus, for its owner, the token functions akin to a digital share in the company. The income streams of the properties are also digitized and linked to the obligation, which differentiates Crowdlitoken from other firms and is therefore the core innovation of the company. This obligation leads to the token holder capturing income streams, such as rental income, as well as increases in the value of a given property. In addition, the CRT token is a digital bond that is linked to a security identification number. The International Securities Identification Number consists of a 12-digit combination of letters and numbers. This feature makes it easier to trade the security on stock exchanges and allows it to be integrated into existing systems, such as banking. In brief, the CRT is the first legally backed digital bond that combines bond and real estate returns (Crowdlitoken, 2022). On a technical level, the CRT is an ERC-20 standard token that is issued on the Ethereum blockchain. Thus, it is a contract with fixed claims (Ankenbrand et al., 2020, p. 21).

AnnA Villa

The AnnA Villa is a building in an affluent area. Here, it is used to illustrate the real-life benefits of tokenization and the concepts that underlie it. In July 2019, this villa, which is worth 6.5 million euro, became the first building to be sold via a blockchain transaction. For this transaction to be possible, the ownership of the building had to be transferred to a public limited company called SAPEB AnnA. Thereafter, ownership rights in the company were linked to the blockchain via a smart contract and divided into 1,000,000 tokens. The smart contract of the token contains all of the property rights that investors have, including shares in rental income. These rights to revenue are also managed and liquidated through the smart contract. The issue involves 1,000,000 tokens, and each token has a nominal value of 6.50 euro. Therefore, the total volume of the issue amounts to 6,500,000 euro. This highlights the benefits of tokenization: without it, an interested party would have raised 6.5 million euro to buy the AnnA Villa, an amount that is beyond the means of the average investor. However, tokenization allows a private investor to participate with a small sum of 6.50 euro (Invao, 2019). Equisafe, which is a digital investment bank, managed the tokenization of AnnA Villa. In August 2019, after the STO had already taken place, Equisafe provided an update on AnnA Villa. The distinction between a classical operation and an operation like AnnA Villa was discussed in the update. Furthermore, they also demonstrated how the rights of the company's property that bought the building are now fully encoded in the blockchain and encrypted on Equisafe's virtual registry. The terms and conditions for purchasing, selling, and exchanging securities, as well as the rights that the securities contain (dividends, voting, etc.), are all included

in each token. Because of blockchain, this information is immune from data leakages, hard to falsify, and always available. Furthermore, the process took only about 30 minutes to complete. Under the traditional approach, it would normally take approximately two months, with costs starting from 20,000 euro (Fries, 2019).

Blockimmo

Blockimmo is a blockchain-based start-up that is on the verge of becoming the leading real estate tokenization marketplace. Blockimmo can offer fractional property investments and ownership because of their regulation-compliant ecosystem. On the Ethereum blockchain, properties are formally and legally represented as security tokens. The tokens that are issued by Blockimmo are also ERC-20 standard tokens. In order to build a cross-border market, Blockimmo collaborate with financial institutions and other strategic partners in several countries. Blockimmo, which is based in Switzerland, is dedicated to making the real estate market more accessible and simpler while providing value to users, one step at a time. As a result, a global real estate market that is significantly more efficient than the contemporary stock market will emerge (Blockimmo, 2022). Blockimmo is also known for tokenizing the first Swiss property in Baar, which made headlines across Switzerland. In the process, over 3,000,000 francs of value in the real estate at Grabenstrasse 3 in Baar were tokenized, which represented 20% of the total value of the property. The tokens were then offered for sale in a special deal to four investors. The Baar property comprises with only 20% of it already been sold which consists of 18 apartments and the Hello World restaurant. The four transactions could be validated on the Blockimmo platform since they were completed on the Ethereum blockchain. Three companies were involved in this Swiss innovation: Blockimmo, Elea Labs, and Swiss Crypto Tokens. Elea Labs provided complete and validated data for the property (Brandenberg, 2019). Elea Labs assumed responsibility for ensuring that each building had its own identity, a “property DNA.” In this way, the property became real: all the data about it was secured and stored in a decentralized manner. Elea Labs' decentralized property DNA allowed customers of the Blockimmo platform to make informed investment decisions on the basis of fully verified property data. Property DNA in the future will make traceable digital due diligence possible. It will also make operations in the real estate industry in general simpler, faster, and more cost-effective (Payoff, 2019). Swiss Crypto Tokens facilitated the transaction through the CryptoFranc, a stable token that is directly tied to the Swiss franc. Its use enabled the risks of cryptocurrency price fluctuations to be avoided (Brandenberg, 2019).

2.2.2 Benefits of the tokenization of physical assets

The tokenization of physical assets, such as real estate, has several advantages. Sazandrishvili (2019) mentioned and summarized six different benefits of asset tokenization. Global access to investments was the first. Those who live in advanced economies find it much easier to access vast investment opportunities. For example, in the USA, 50% of households hold stock in publicly traded companies. Conversely, in transitional and developing economies, only 1% or 2% of the population invest in the stock market. The tokenization of assets would make it possible for investors to access opportunities across the globe due to the accessibility of fractional ownership and the elimination of intermediaries (Sazandrishvili, 2020, pp. 69-70). In a perfect world, tokenization would also enable trades to be executed at all hours: all that a buyer would need in order to buy an illiquid asset would be an ERC-20-compatible wallet (Markheim & Berentsen, 2021, p. 72). The second benefit has to do with liquidity, which is also associated with fractionalized ownership and universal access to investment opportunities. Allowing smaller investors to access more markets results from the ease with which digital assets can be traded.

It also creates convenient means of exchanging, transferring, and liquidating assets that are both easy to use and secure. This convenience will attract more small investors, which will make the market more volatile (Sazandrishvili, 2020, pp. 69-70). Given that the issuer can trade the tokens on a secondary market, which leads to higher liquidity, expanding market coverage can be seen as a significant benefit of the tokenization of physical assets. This opportunity offers advantages to investors and to sellers (Laurent et al., 2018, p. 2). The third benefit is the reduced need for intermediaries. Tokens can be traded directly between buyers and sellers, without the involvement of third parties such as banks, lawyers, or other middlemen. The transaction experience becomes more satisfactory for market players. However, intermediaries may occasionally be needed to verify the existence and condition of the underlying asset. This type of intermediation can develop further in order to leverage the benefits of current blockchain technology, which would still be more efficient than the use of traditional intermediaries that affects the process in the long run. More rapid transactions are the fourth benefit. Assets can change their owners at the speed of thought because there is no need to sign papers, to wait for waiting periods to lapse, and such like. The transactions occur in real time and can be accessed from any location through a web browser or a mobile device. The digitalized process thus entails little or no human intervention where common violation of security can be attained (Sazandrishvili, 2020, pp. 69-70). Laurent, Chollet, Burke, and Seers (2018) elaborated on this benefit, adding that tokenization makes transactions not only more rapid but also cheaper due to the use of smart contracts and the resultant automation of the exchange process. Automation leads, among others, to a reduction in the administrative burden, which ultimately lowers transaction fees (Laurent et al., 2018, p. 2). The fifth benefit is transparency. As far as transparency is concerned, the image of blockchain has been tarnished—anonymous cryptocurrencies are associated with criminals, terrorists, and crime. In reality, blockchain is one of the most transparent financial systems in history. Every single transaction can be publicly observed or traded on digital ledgers, which makes the system more transparent than that of traditional banking. The last benefit is immutability. In this context, the term “immutability” refers to the fact that information is almost impossible to alter or delete after a block is confirmed and added to a blockchain. The records of the transactions in question are replicated on computers around the globe, which means that when a hostile party attempts to alter a transaction, all other computer operators (miners) recognize the behavior and block the modification. Not only does this process make transaction history transparent, but it also guarantees its security (Sazandrishvili, 2020, pp. 69-70).

However, these advantages should be analyzed with care. The technical process of tokenizing physical real estate is considered simple, but its practical benefits over conventional forms of financing are not apparent. Regulatory deficits and uncertainties mean that most advantages are theoretical in nature and not yet feasible in practice (Markheim & Berentsen, 2021, p. 59).

2.3 Tokenization of virtual real estate

Tokenization is a broad term that may be used in various contexts (Appelmans, 2022, p. 1). Thus, NFTs are a part of tokenization. As mentioned, NFTs are explained in greater detail in the chapter 2.1.6 on understanding tokens, which is of central importance for virtual real estate (Appelmans, 2022, p. 6). The authors, Heines et al. (2021), provide eight different archetypical application cases of tokenization. NFTs, a subset of tokenization, fall into the category of “digital sovereignty” among these. They explained that this type of token enables users to own, manage and control assets such as virtual real estate in a decentralized platform or world (Heines et al., 2021, pp. 9-10). Virtual real estate is becoming an increasingly important phenomenon because the land rush of the 21st century is unfolding in the digital

environment, that is, in the metaverse, rather than on physical land. Therefore, digital currencies and assets are gaining popularity. In addition, the virtual universe is expanding. Early investors have already invested large sums of money in virtual real estate. However, this development is associated with high risk—the digital market is volatile (Borgenicht, 2022, p. 1). Land and buildings within virtual spaces in which individuals spend a considerable proportion of their spare time are referred to as “virtual real estate.” Many such properties can be obtained by exchanging cryptocurrency for the associated NFTs. Depending on the specific type of virtual real estate, these properties may be associated with monetary rewards (Heidorn, 2022). Virtual real estate and NFTs are more than just gimmicks. These virtual worlds attract many users who are willing to spend a significant amount of money on and time in the metaverse. The phenomena of NFTs and the metaverse are described in greater detail in the next two sections. Initial empirical studies of the pricing of virtual real estate and NFTs have shown that several factors affect the price mechanism. Investors are willing to pay more for virtual real estate if it is located near a city center, a plaza, a main street, or a business district. These findings overlap with observations from the real world. However, there are certain differences in the virtual domain. For example, avatars do not incur transportation charges when moving around the metaverse, and land is used primarily for commercial purposes. As a result, commute and supply-chain factors have no bearing on location selection. Instead, the choices of investors are shaped by expectations about visitor density (Goldberg et al., 2022, pp. 31-32).

2.3.1 The NFT phenomenon

NFT market volume was between 12 and 24 billion dollars in 2021. The Collins Dictionary even announced "NFT" as its word of the year for 2021. In a distributed blockchain database, an NFT technically represents a data entry that records an ownership agreement. As noted previously, these entries are managed through a smart contract. There is 97% of NFT smart contracts are now processing on the Ethereum blockchain. Unlike the tokenization of physical real estate, NFTs use the ERC-721 technical standard (Baytaş et al., 2022, p. 1). The Ethereum blockchain was foundational for the implementation of NFTs. The first standard token that was used for NFTs on the Ethereum network was the ERC-721 token. However, as the popularity of NFTs grew, the standard improved and developed in different blockchains. The first standard for NFTs, ERC-721, which was developed in Ethereum, is free and open source. A smart contract should implement the ERC-721 interface in order to transmit and manage NFTs. In addition, each ERC-721 token has a unique property and a different token ID. This standard token contains information about the owner, a list of authorized addresses, a function that enables the transfer of tokens from owners to buyers, and many other useful features. Smart contracts in ERC-721 can combine tokens with the same configuration, but each token has its own set of attributes. Therefore, FTs are not supported by this standard. The newer ERC-1155 is an Ethereum standard that was designed by Enjin. It has more features than the ERC-721 and allows FTs, NFTs, and SFTs. ERC-1155 makes it possible to program or transmit multiple NFTs through the same smart contract. Therefore, it can manage multiple token types (Bamakan et al., 2022, p. 3).

The existence of ERC-721 NFTs, such as Beeple’s digital artwork that was sold for 69,000,000 dollars or Twitter CEO Jack Dorsey's first tweet that was auctioned for 2,900,000 dollars, shows that NTFs are gaining attention and becoming mainstream (Ante, 2021, p. 1). One of the most substantial public breakthroughs of blockchain technology is the rapid growth of the NFT market seen by its popularity. The hype around NFTs goes far beyond art, videos, music, and such like. It has also reached virtual worlds, and virtual real estate is being tokenized. One example of a virtual world is the Decentraland project,

which is based on blockchain and often referred to as a project within the metaverse. The metaverse is described in the next section. In the world of Decentraland, there are a total of 90,601 plots, of which 43,689 are for private use. These private plots can be traded freely. Virtual real estate can be purchased in the form of NFTs through the cryptocurrency MANA. Each transaction or ownership change is recorded in an Ethereum smart contract. When virtual property is purchased in Decentraland, it can be employed as desired. For example, it can be used to host games, events, or shops that offer virtual or real goods. Visitors can also listen to exclusive concerts—the possibilities are endless (Dowling, 2021, p.1). Plots in Decentraland are priced differentially. Like in the real world, the location of a plot is an important determinant of price. Virtual real estate in locations that are seen by more individuals commands higher prices. For example, investors are willing to pay more for properties that are located in city centers (Kräussl & Tugnetti, 2022, p. 20).

2.3.2 Metaverse

Well-known technology companies have already embraced the metaverse trend, as have consumer brands such as Gucci or Coca-Cola. These companies sell NFTs on metaverse platforms such as Decentraland. Thus, the term “metaverse” has generated much hype and helped NFTs, as well as their uses in virtual real estate, to become more popular (Kim, 2021, p. 141). However, NFTs are often pushed to the background—the metaverse has become the new buzzword, and everything around it is forgotten. Nevertheless, NFTs are essential to the metaverse. All products and services in the metaverse are, in essence, NFTs. Accordingly, virtual real estate plays an important role (Kiong, 2022, p. 29).

What exactly is this metaverse, and what does it do? The term “metaverse” first appeared in a science fiction novel in 1992. The novel was called *Snow Crash*, and it was written by the Neal Stephenson. His metaverse was a three-dimensional virtual world. “Meta” stands for “virtual” and “abstract,” and the “verse” stands for “universe.” The metaverse is therefore a virtual concept whereby users interact with each other through avatars that they create in order to participate in or to reproduce real life in a metaphorical setting that is free of temporal and spatial limitations (Sina Osivand, 2021, p. 125).

After the first attempts to define a metaverse, which occurred decades ago, the metaverse concept came to be applied in computer games, which contain virtual worlds. Gaming companies tried to offer users a unique experience in those worlds. Attempts were made to create numerous metaverses, one of the most famous being Second Life. Second Life was developed by Linden Lab in 2003. This virtual world promised users a second alternative world in which they could create their own avatars along with all their features. The only limitation was the imagination of the individual. Socializing, buying real estate, designing, and acting out fictions were just some of the possibilities that this world offered. One could even receive a university education. Second Life created its own economy, using the Linden dollar as currency. The Swedish government even opened an embassy in this virtual world. Today, attempts to create metaverses continue. The examples in gaming include Sandbox, Roblox, and Fortnite. However, the metaverse topic, along with NFT hype, returned to the spotlight when Mark Zuckerberg issued a press release in which he reported that he would be changing the name of his company to Meta. This development gave a new meaning to the metaverse and led many to explore it. This proposition is also evidenced by Google Trend reports. In October 2021, a sharp increase in searches for the term “metaverse” was noted. Given the importance of social media, particularly among Generation Z, investments in the metaverse are influencing future technologies. According to Zuckerberg, the metaverse is the next big thing on the internet (Narin, 2021, p. 17).

Mystakidis (2022) essayed a contemporary definition and described the metaverse as a post-real universe that unfolds in a perpetual and persistent multiuser environment and in which physical reality merges with digital virtuality. It is built on the convergence of technologies like virtual reality (VR) and augmented reality (AR), which allow for multisensory interactions with virtual environments, digital items, and people. In addition, the author defined the metaverse as follows:

Hence, the Metaverse is an interconnected web of social, networked immersive environments in persistent multiuser platforms. It enables seamless embodied user communication in real-time and dynamic interactions with digital artifacts. Its first iteration was a web of virtual worlds where avatars were able to teleport among them. The contemporary iteration of the Metaverse features social, immersive VR platforms compatible with massive multiplayer online video games, open game worlds and AR collaborative spaces (Mystakidis, 2022, p. 486).

Definitions of this kind should be treated with caution—experts have not yet agreed on a standardized interpretation of the metaverse. Although many speak of a metaverse, the concept is often misunderstood. The metaverse is not necessarily about decentralized worlds, and it can be centralized. No industry consensus has emerged because the metaverse concept is too new and broad. Although there are still no clear definitions and perspectives vary, many experts agree that the interpretation of the metaverse should account for various attributes. Persistence is one of them—a metaverse must always exist and never restart, pause, or terminate (Kiong, 2022, pp. 22-24).

Beyond social media and commerce, gaming is considered the most successful use case of the metaverse. Developments from past years confirm this observation. The decentralized metaverse games that have emerged are combined with NFTs. By creating new possibilities, they are displacing centralized worlds—players are moving to the decentralized gaming metaverses. The Sandbox Metaverse may serve as an example. Within that game, players can build, own, and monetize their game experiences by using a utility token called SAND. Users who own SAND tokens can vote on important ecosystem decisions. The Sandbox environment offers various options. For examples, players can create NFTs and sell them on a marketplace if needed. The Sandbox virtual world contains 166,464 LANDS, that is, physical spaces that are owned by players and can be monetized. In this case, virtual real estate is represented by LANDS as NFTs. The purchase of such virtual plots allows players to own a part of the metaverse and to host content (Kiong, 2022, pp. 31-33).

2.3.3 Challenges NFTs

NFTs have a great influence in the metaverse, as they enable virtual worlds to trade assets and virtual properties such as art, names, wearables, and virtual real estate. Thus, it is also possible for users to make profits from this virtual economy. Virtual real estate in these worlds, such as in Decentraland, can be rented out to other people (Q. Wang et al., 2021, p. 12). However, in order for this concept or development to become more mature, several challenges still need to be addressed and solved. The authors Wang et al. (2021) mentioned four major challenges.

The usability challenges are addressed first, which can be divided into two points and have a direct influence on the user experience. The issue of slow confirmation and high gas prices are addressed. Transactions are slow since the NFT systems are coupled to specific blockchain platforms, which are not yet matured. So, it takes a longer time until a transaction is confirmed. In addition, NFT-related transactions are associated with high costs, as smart contracts are associated with expensive resources (Q. Wang et al., 2021, 13). This is also due to the increasing public interest in NFTs in recent years.

Since mainly NFTs are traded on the Ethereum network, this results in high gas fees (J. W. Kim et al., 2022, p. 64).

Moreover, there are challenges with regard to security and privacy. Various risks of losing this link or malicious parties attempting to exploit this vulnerability exists since data is stored off-chain but refers to on-chain tags. Consequently, it is possible that this system's flaws will result in the NFT data becoming inaccessible. But there are also deficits in terms of privacy and anonymity. This point is still very unexplored in relation to NFTs. Since most NFT transactions take place on the Ethereum platform, strict anonymity or privacy are not provided. Existing privacy solutions cannot be applied due to the complex cryptography of NFTs and require new approaches in this area (Q. Wang et al., 2021, pp. 13-14). Regarding security, attacks on decentralized financial protocols and vulnerabilities in the NFTs' smart contracts are currently being investigated. The marketplaces where NFTs are traded have hardly been the subject of investigations (Das et al., 2021, p. 1).

NFT-related regulatory issues are also starting to surface. In order to properly regulate this emerging technology and its corresponding market, various areas such as cross-border transactions need to be addressed. It presents governance issues, such as Know your Customer (KYC) data and others. There is also a need for regulation on the specifics of how NFTs should be taxed and when a higher tax bracket should be applied, which relates to the problem of taxable property issues. Finally, there are challenges related to extensibility. These challenges can be divided into NFT interoperability and updatable NFTs. The former deals with the fact that the existing NFT ecosystems can only be traded within this ecosystem and thus cannot access other ecosystems or networks. This imposes an interoperability constraint that prevents widespread adoption. The generic paradigm requires that a new version follow certain rules, including permitting specific aggressive behaviors and continuing to operate online while updates are being produced. NFT systems strongly rely on and maintain consistency with their underlying platforms. The most crucial logic is still kept on-chain even though the data are frequently stored in distinct components. Updating the system properly and adding enhancements will be required (Q. Wang et al., 2021, pp. 14-15).

3 Methodology

The following chapter describes the methodological procedures that were followed in this master's thesis. The research design, which includes a literature review and qualitative interviews, is outlined. Furthermore, the chapter describes the subject of the qualitative interviews and the manner in which they were conducted. In addition, the structure of the interviews is overviewed, as is the selection of interviewees. The analysis of the interviews was conducted in line with the work of Mayring (2015, pp. 70-72). Finally, the results of the application of the empirical research method are evaluated and presented.

3.1 Literature review

On the topic of the tokenization of real estate as it relates to physical and virtual assets, a thorough literature search was undertaken. The topics that precede this one and are crucial to comprehending it were fully defined. Blockchain was examined thoroughly. One of the use cases — tokenization, was considered in general. In addition, the transition from physical real estate to digital real estate was studied. This theoretical foundation was meant to serve as a link between the parameters for the study and the research topic. The idea that follows exemplifies the standards under which the literature review was done.

Creswell (2014) stated that an author should identify not only the literature that they intend to use but also the manner in which they intend to use it. He wrote that a satisfactory research project should be easy to read and understand. He favored the use of a short title of no more than 12 words in order to sharpen the focus on the topic (Creswell, 2014, pp. 24-25). Neumann adds that a literature review provides an overview and summarize previous studies (Neuman, 2014, p. 30). The author Cooper (1982) describes that there are multiple types of literature research. According to him, literature reviews can be integrative, whereby the researcher summarizes specific topics from the literature. They can also criticize existing work or build connections between topics. In addition, literature reviews can facilitate the identification of a central problem in a research domain (Cooper, 1986, p. 17). The identification of existing and significant literature is a necessary feature of any scientific project and is therefore unthinkable. A detailed and effective review creates a solid foundation for knowledge growth and identifies research gaps (Webster & Watson, 2002, p. 13).

Creswell (2014) developed a seven-step model. The model is used to support the development of a literature review. According to him, the purpose of literature research is to identify and collate studies on a topic. There are many ways to attain this goal. However, researchers usually proceed systematically. He thus recommended the following seven steps (Creswell, 2014, p. 25):

1. The first step has to do with selecting keywords that can enable the identification of relevant publications in university and college libraries. These keywords may emerge from preliminary research. In the instant case, the keywords that were used included “blockchain,” “tokenization,” “smart contracts,” and “real estate.”
2. In the second step, these keywords are used to search for relevant articles in library catalogues. The focus is mainly on books and journals that are related to the topic. The databases that are recommended for social science researchers include the Educational Resources Information Centre (ERIC), EB-SO, ProQuest, Google Scholar, PsycINFO, and others.
3. The third step entails finding approximately 50 research papers, preferably from books or journals, that are related to the topic. Books and journals are recommended because they are easier

to find. Furthermore, it is important to ensure that the literature that one finds can be retrieved at any time. This is also necessary to ensure the availability of physical literature.

4. In the fourth step, all the literature that has been identified is skimmed in order to create an overview. In this step, important information from each publication is noted. The purpose of the step is to determine whether the books and journals that have been found are useful for the understanding of the topic.
5. The fifth step is about visualization. Visualization takes the form of a literature map or, to be precise, a visual image or picture of a thematic subgroup that is related to the topic. It should illustrate what specific studies contribute to the research domain.
6. In the sixth step, the researcher composes summaries of the most relevant papers. These summaries, which should reflect the use of an appropriate style of citation, are included in the final literature review.
7. In the seventh step, the final literature review is written. Requirements, such as structuring the exposition so that it is comprehensible, must be fulfilled. Literature research concludes with the composition of a literature review, which is a summary of the most important topics and works. In the last stage, the researcher should ensure that the literature research is sufficient. If gaps are present, it should be extended (Creswell, 2014, pp. 64-65).

3.2 Guided interview

It is difficult to conceive of empirical research without the interview as a scientific method of data collection. Interviews are a qualitative research method. Guided interviews are popular (Loosen, 2016, p. 139). Compared to other qualitative interviews, guided interviews entail a higher degree of standardization (Prochazka, 2020, p. 121). Interviews in general and guided interviews in particular are more respectable than other forms of qualitative research because they are used more frequently (Diekmann, 2018, p. 435). Predefined guidelines specify the topics and questions that are discussed with the interviewees. Since the same questions are asked, the responses can be compared. Guided interviews are also flexible. The order of the questions can be changed at will, and some questions can be skipped. In addition, it is possible to ask questions that do not feature in the guidelines. New aspects of the topic can be investigated in this way. Finally, the goal of a guided interview is to conduct an open and free-flowing conversation (Prochazka, 2020, p. 121). Despite their flexibility, guided interviews are structured clearly (Stigler, 2012, p. 141).

3.3 Expert interview

An expert interview is a special form of guided interview. This particular variant is used when the subject is to be interviewed as an expert and not in a private capacity. The questioning follows a guide, which contains open questions. In this particular type of interview, the questions serve mainly as an reference points that ensure that the fundamental issues remain in focus during the course of the interview and that the interviewee does not digress too much from the subject matter (Kaiser, 2014, pp. 1-3). Qualitative research, which includes interviews, is useful when the domain that is being studied is still largely unexplored. Thus, qualitative research is suitable for relatively new topics on which there is a lack of theory due to a dearth of prior research. Therefore, it is necessary to explore and describe a

particular phenomenon, which can lead to the creation of new theories (Morse, 1991, p. 120). The interviews can be conducted in different forms, such as in person (face to face), by phoneme, or online (Skype). Each format has advantages and disadvantages (Loosen, 2016, p. 145).

An expert interview, as its name implies, necessitates the presence of an expert. It is crucial to know whether someone is regarded an expert in a given situation. In common speech, experts are defined as specialists or professionals in a specific area. In other words, they are individuals who possess special knowledge. Thus, an individual becomes an expert if they possess a clear knowledge advantage in relation to the research questions. In addition, experts have privileged access to information about groups of individuals and about decision-making processes (Liebold & Trinczek, 2009, pp. 33-35). A more efficient and focused method of data collection than, for example, observation or systematic quantitative surveys is speaking with such experts during the exploratory stage of a project. Furthermore, interviewing an expert can accelerate the data collection process, particularly when experts are treated as crystallization points for having relevant inside information and are being asked on behalf of a broader group of stakeholders. In addition, expert interviews allow access to a specific social field, which may otherwise be difficult or even impossible. If the expert is eager to cooperate and holds a key position in the organization, there might be further opportunities for the researcher to penetrate the field. An expert may refer to potential interviewees who are experts in a particular domain. Such assistance makes it easier for the researcher to reach a larger group of experts (Bogner et al., 2009, p. 2).

The process of conducting an expert interview can be divided into four steps. The first step entails the creation of guidelines. In this step, it is important that the interviewer address the subject in depth, which allows the conversation to be conducted on equal terms (Qu & Dumay, 2011, p. 239). The second step has to do with the targeted selection of experts. For the researcher to determine whether an individual meets the criteria for expertise, which is what allows relevant data to be obtained, background information must be gathered (Flick, 2016, p. 294). The third step is data collection, that is, conducting the interviews. In this context, the significance of familiarizing oneself thoroughly with the topic must be emphasized once more. The final step entails evaluating and analyzing the interviews (Kaiser, 2014, pp. 6-7).

This theory was put into practice in this master's thesis. As a result, the guidelines were structured well. It was necessary to conduct a comprehensive literature review first. It was followed by the composition of the theoretical part of this thesis, which yielded high-quality guidelines that could be used to speak to the experts on equal terms. The guidelines are divided into three sections. The first section is about opening the interview effectively and covers general questions, such as the experts' first contact with blockchain and the areas in which they see the greatest potential for blockchain technology. This allows for a more relaxed introduction to the topic of blockchain, without any direct reference to the tokenization of real estate. Furthermore, this first section enables the researcher to familiarize themselves with the experts better and to obtain the first essential data. The next section of the guidelines concerns the tokenization of physical real estate. Various facets of this topic are explored because it is one of the fundamental aspects of the research. The guidelines cover the difficulties, opportunities, and regulatory implications of tokenizing physical real estate. The experts' experiences with this activity are elicited. Those experiences occupy a central position, and assertions are made on that basis. The goal of the section is to analyze the topic from different angles and to generate insights on various levels. The third section of the interview design focuses on the tokenization of virtual real estate in order to answer the research question appropriately and to compare physical and virtual real estate in the context of tokenization. Experience is at the forefront once more. The goal is to refer to the NFT phenomenon to identify important facts about virtual real estate. The expert's personal viewpoint also plays a significant role,

allowing for multiple perspectives to be explored on several levels. For example, the key points of difference between physical and virtual real estate are discussed. As a result, the novel phenomenon under observation can be described, and new insights can be gained.

3.4 Pretest

If a questionnaire is created with little knowledge about a subject, errors are bound to occur if it is not tested (Hague, 1987, p. 170). However, in this master's thesis, an extensive theoretical examination was conducted before the questionnaires and the guideline were designed in order to achieve the best possible results. The importance and necessity of pretesting is strongly emphasized in the literature, regardless of the level of theoretical engagement with a research topic. Pretesting is recommended as a tool in all contributions to the extant literature, especially when it comes to novel research topics (Reynolds et al., 1993, pp. 1-2). Kromrey and Strübling (2009) also stressed the importance of pretesting, particularly in relation to quality assurance and successful interviewing (Kromrey & Strübling, 2009, p. 384). Two main elements of a design are usually pretested. The first concerns individual questions; the second is about the overall design of the interviews. The former is considered a more obvious subject for pretesting (Oppenheim, 1966, p. 28). The pretest should extend beyond individual questions and ensuring that no questions are missing, that the questions are significant to the study, and that no questions are redundant. The logical sequence and the flow of the interview should also be tested and verified (Reynolds et al., 1993, pp. 2-3). Two interviews were conducted in advance, as a pretest, in order to ensure the appropriateness of the data collection method and to avoid complications during the interviews with the experts. The two pretests were executed with the aim of identifying and eliminating potential problems as early as possible. This pretest enabled the quick identification of weak points and the subsequent revision of the interview guideline (Schnell, Hill, & Esser, 2013, p. 3).

Since the research topic, especially the recent development of NFTs, is relatively new and highly specific, it was not easy to find subjects for an effective pretest. A fellow student who had accumulated knowledge about blockchain technology for several years and had completed a certificate course on blockchain and fintech was recruited. This made it possible to conduct two pretests two weeks before the first expert interview. Sufficient time was left for deficiencies to be corrected without pressure. The pretest entailed checking for various issues, such as the comprehensibility and formulation of the questions. In addition, the flow of the interview was monitored, and a clear structure and a common thread running through the questions were sought. The process also indicated how much time each interview would take. The first session highlighted the importance of pretesting—several questions had to be rewritten, removed, or supplemented. For example, many questions were speculative. Such speculative questions yield speculative answers and do not contribute significantly to scientific rigor. Therefore, the questions were geared to knowledge and experience. Furthermore, there were problems with the design and the flow of the interview process, which meant that the order of the questions had to change. The findings from the test run and the feedback from the pretest subject were analyzed and implemented. After all changes and improvements had been implemented, a second test run was carried out. This second pretest served to ensure that the feedback had been integrated correctly into the design and to identify errors that had been overlooked. That test ran smoothly: every single question could be answered clearly, and there was an obvious connection between the questions. Since no problems were identified, the final phase was carried out. That phase had to do with the output of the interviews. The goal of interviews is to elicit data that can be used to answer a research question. Thus, the data from

the two test procedures were analyzed. It emerged from the evaluation that the questions had been chosen optimally in order to obtain a result that would be meaningful and adequate in terms of content, thus enabling a satisfactory evaluation of the data. In summary, the importance of pretesting, which has been mentioned several times in the literature, was confirmed—the pretest was indispensable for the successful execution of this empirical study.

3.5 Conducting the interviews

The seven interviews were conducted orally and in German. The interviewees were native German speakers. Therefore, no relevant information was lost due to language difficulties. Six of the seven interviews were conducted via Zoom, and one was conducted in person. A timeframe of one month was set for the interviews, which took place between 1.5.2022 and 31.5.2022. The experts were informed about the topic in advance. Before the interviews started, they were asked whether the output should be anonymized. In addition, each interviewee agreed that their interviews could be recorded. Questions about experiential knowledge were asked in the course of the interviews.

3.6 Selection of experts

The selection of interviewees, who were experts, raised the question of how many interviewees would be needed to provide a professional and scientific answer to the research question, which is the main focus of this master's thesis. Baker and Edwards (2012) did not specify a number of experts that are required for qualitative interviews. There are various external elements that influence how many experts are ultimately interviewed. The amount of time that is available to execute a research project and the difficulty of recruiting experts on certain topics should be considered. Although considerations that are internal to the project should take precedence, external considerations frequently have a significant effect on the number of interviewees (Baker et al., 2012, p. 6). Saunders and Townsend (2016) wrote that the number of interviewees can vary considerably depending on the goal of a work. However, an appropriate balance should be found—if there are too few experts, it becomes almost impossible to evaluate the data. Speaking to too many experts can also be problematic (Saunders & Townsend, 2016, p. 845). Therefore, the literature contains a variety of recommendations and viewpoints. Many articles cover this topic, with recommended sample sizes ranging from 5 to 50. The quality of the data, and thus the amount of relevant data that can be acquired from the participants, is an important factor in determining the optimal number of interviewees (Dworkin, 2012, pp. 1,319-1,320).

In this master's thesis, the objective was to interview between five and 10 experts. Considerable care was taken to ensure that the experts would be suitable for the research topic and would thus be in a position to provide high-quality and relevant information. Since, as mentioned above, answering the research question is of central importance to this master's thesis, only qualified interviewees were sought. Finding the desired experts was not easy because the topic is new and specific. For the differences between the tokenization of physical and virtual real estate to be shown, it was necessary to recruit experts who are active and experienced in the field of blockchain. In addition, the experts were expected to possess knowledge about real estate and to have experience or academic knowledge of that sector in the context of blockchain. The experts were further expected to have interests beyond their activities, in that they would also follow other blockchain developments, such as the emergence of NFTs and virtual real estate. This expectation reflected the need to examine different perspectives on the crypto space.

The backgrounds of the experts, including their biographies, were examined, and suitable interview partners were contacted by telephone and e-mail.

In total, seven interviews were conducted. The interviewees met the aforementioned criteria and were thus able to supply relevant information. One of the interviewees did not specialize in blockchain technology. This expert was a traditional real estate agent and was chosen deliberately in order to learn more about the traditional real estate market and its processes. In addition, it was possible to speak to them about the possibility of existing issues in that market being solved by blockchain technology and about awareness of blockchain technology among the public.

Table 2 describes the backgrounds of the interviewees and their designations (“IP1,” “IP2,” and so on). For ease of exposition, the findings from the qualitative interviews are presented using these designations rather than the names of the interviewees. The anonymization of the responses enabled the author to strike a balance between the need to protect the identities of the participants and that to maintain data quality and reliability.

Designation	Occupation	Degree	Company
IP1	CTO	Master’s degree in marketing, communication, and management	Blockchain company
IP2	CEO	MBA	Blockchain Company
IP3	CEO	Degree in design management	Blockchain company
IP4	CEO	PhD in finance	Blockchain company
IP5	Private investor and trader	Master’s degree in event management	Freelancer
IP6	Project Manager	Master's degree in business and economics	Financial services company
IP7	CEO	Degree as real estate appraiser	Real estate agency

Table 2: Profile information of interview experts

Source: Own illustration

What follows is a description of the interview partners (IPs). Additional statements from the interviews are used to describe the IPs. Specifically, reference is made to the general question about the manner in which the IPs first came into contact with blockchain.

IP1

IP1, who was working as a Chief Technology Officer, was in charge of IT and blockchain connectivity and had graduated with a master's degree in marketing, services, and communication management. IP1 worked as a consultant and project manager on software projects for banks before joining a real estate company that employs tokenization to enable investments in digital real estate. IP1 had been specializing in blockchain technology since 2015 and had held different roles in the nascent blockchain industry,

most recently as the CTO of a company that tokenizes financial products. IP1 first learned about blockchain technology from news articles and subsequently became interested in the subject. From a strictly ideological standpoint, IP1's interest was piqued by the idea of non-fiat currencies, such as Bitcoin, that are not overseen by a central authority.

IP2

IP2, the CEO of a Swiss cryptofinance and technology company since July 2019, had obtained a Master of Business Administration (MBA) degree. Before assuming that role, IP2 had worked as Head of Blockchain and PSP at a bank. Furthermore, IP2 had successfully implemented a corporate strategy in the field of blockchain banking and had also established its Blockchain and Payment Service Provider department. Thus, IP2 has driven the establishment and development of a blockchain banking service. IP2 first learned about Bitcoin from a university lecturer in 2012 but did not take it seriously at the time. Later, IP2 rediscovered blockchain technology through work. As a result, IP2 became interested in the subject and gained additional knowledge.

IP3

IP3, founder and Managing Director of a blockchain consulting and solution development company and a real estate investment platform, studied design management. In the 15 years before the interview, IP3 had founded and built several IT and real estate businesses as an entrepreneur and active investor. IP3's most recent company was a blockchain-based real estate tokenization platform. IP3 had always been digitally active and had therefore spent much of their time on IT-related activities, such as programming websites. IP3's first foray into the real estate industry was also entirely digital. IP3's activities were such that it was only a matter of time before they came across blockchain technology. Thus, IP3 became involved with cryptocurrency technology early. Instead of focusing on price speculation, they accumulated considerable know-how in the area.

IP4

IP4 had considerable expertise in bridging the gap between traditional finance and emerging technologies, and they had been involved in various digital asset and tokenization projects since 2017. IP4 was instrumental in the successful launch of the first European provider of digital financial investments for real estate, among other things. Previously, IP4 had been the managing partner of a consulting boutique that specializes in quantitative finance, and they had worked for the world's leading providers of reinsurance, insurance, and other forms of insurance-based risk transfer companies in various countries. IP4 had obtained a PhD in asset management.

IP5

IP5 previously worked in the event management industry. When IP5 first heard about Bitcoin and blockchain technology from friends and acquaintances in 2012, they were still fairly ignorant of the subject. However, this changed in 2017, when a friend advised IP5 to transfer money to a cryptocurrency exchange. Along with their profession, IP5 developed a substantial interest in investing and trading, and they were working as a freelance crypto trader at the time of the interview. IP5 started accumulating trading profits from short-term fluctuations in the market value of cryptocurrencies, altcoins, and value tokens. Since then, IP5 had been studying blockchain and keeping a close eye on developments.

IP6

IP6 had been working as a product manager at a financial institution that specializes in private banking since 2019. IP6 graduated with a master's degree in business and economics. They first learned about

blockchain at a guest lecture at university, roughly seven years ago. Their knowledge was still relatively superficial at that time, but it evolved as a result of their activities at work. Most of IP6's knowledge was gained on the job, and they had participated actively in blockchain-based projects.

IP7

IP7 was the owner and Managing Director of a traditional real estate agency. They had been in that position since 1991. IP7 had obtained a degree as a real estate appraiser and was also a member of an appraisal commission. In addition to the sale of real estate, IP7's responsibilities included real estate evaluation according to the DCF and MEV methods, project development, and consulting on real estate management and financing. Despite being a traditional real estate marketer rather than a blockchain expert, IP7 offered interesting insights into the similarities between traditional and virtual real estate. Nonetheless, blockchain was not a new phenomenon for IP7, who had already invested in the technology.

3.7 Content analysis

There is no single correct approach to evaluating qualitative expert interviews through content analysis—different qualitative evaluation strategies can be used. Summary content analysis is a popular method that is considered efficient (Wassermann, 2015, p. 61).

Stuckey (2014), drawing on his experience, wrote that transcription quality can affect the output of an analysis. He therefore emphasized the importance of the former. He saw the transcription of an interview as the first and most important step in qualitative content analysis (Stuckey, 2014, p. 7). A simple transcription method, in line with the works of Dresing and Pehl (2018), was selected. The recordings were transcribed verbatim in a simplified form. No summary was transcribed. However, each spoken word was taken into account. The rules state that filler words, such as "mhm" and "ähm," as well as stuttering and other sentence breaks, do not have to be transcribed. In addition, dialects were translated into High German. Appropriate punctuation was used, which led to better readability. This procedure was designed to ensure that the transcript would be easy to understand (Dresing & Pehl, 2018, pp. 21-22).

In this master's thesis, transcription was carried out by using the procedure and the rules that were described above. This yielded 30 pages of transcription. The content analysis technique was used to evaluate the data. It is explained below. After the first step of the content analysis, namely transcription, had concluded, seven further steps were taken, in line with Mayring (2015).

It must first be determined whether it is better to adopt an inductive or a deductive approach. Both approaches entail comparable preparatory stages, although they are fundamentally different. A deductive method is based on previous theories or models, that is, on a pre-existing theoretical framework. If there is insufficient prior knowledge about a phenomenon, the inductive technique is recommended. In inductive content analysis, categories crystallize from the data (Elo & Kyngäs, 2008, p. 109). The present master thesis examines a very recent topic, virtual real estate and the use of NFTs, which is one of the reasons why the experts were primarily interviewed about their personal experiences. Therefore, the inductive approach was chosen.

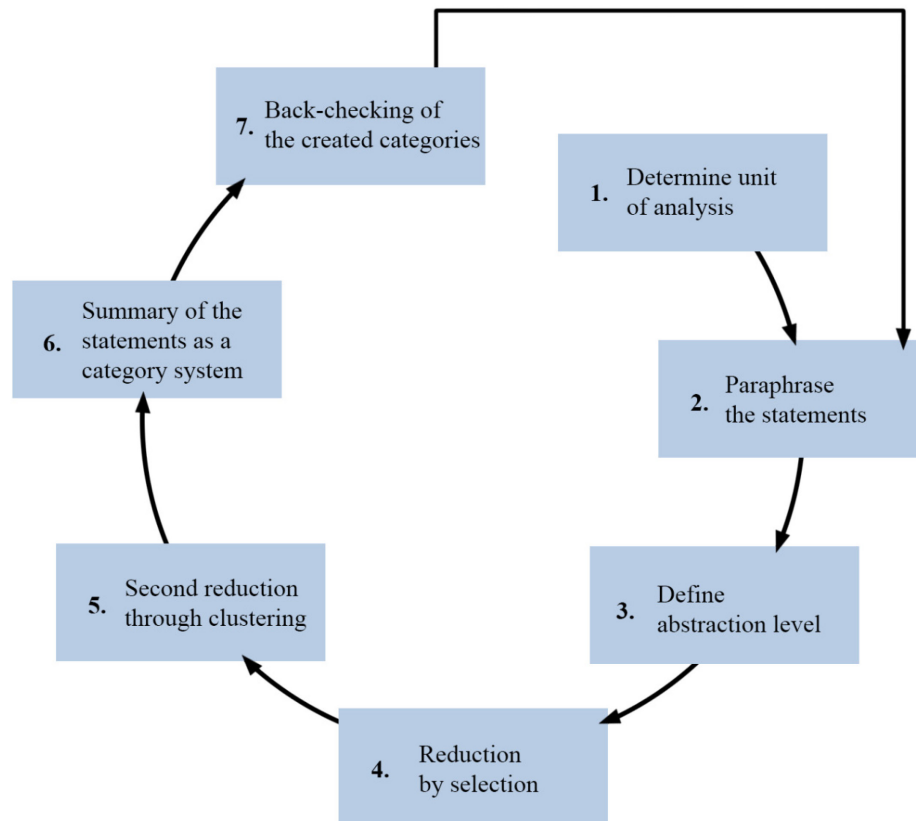


Figure 3: Seven steps of inductive category creation

Source: Mayring (2015, p. 70)

Once an inductive approach has been chosen, the seven steps in Figure 3 can be performed. The determination of the units of analysis is the first stage. The answers to a question are used to determine which part of the interview should be evaluated. The single coding units are then sorted by meaning and summarized, or, in the terms of Figure 3, "paraphrased." In other words, unnecessary material is eliminated. Thus, a carefully recorded central statement becomes the focal point of the work. The documented claims are subjected to abstraction in the third step, which leads to the generalization of claims. A first reduction by selection occurs in the fourth phase. Words that do not have any genuine meaning are removed. Paraphrases that have the same meaning are summarized and rewritten. Several interconnected paraphrases that are typically dispersed throughout the material are summarized and reproduced by a new statement in the second reduction stage. The statements that have been transcribed are collected in the sixth stage, when a classification scheme is formed. It is critical to review all claims separately and then to integrate them into a category while using the inductive approach. In the final phase, the category system is tested to see if it is still correlated to the source material and if the reduced statements are still representative. In the event that this is not the case, the degree of abstraction can be changed, and the process must start over at the second step (Mayring, 2015, pp. 70-72). The scheme that is presented in Figure 3 was used for this master's thesis. All of the steps were carried out in an identical manner.

4 Empirical Results and Evaluation

This chapter presents the results that were obtained from the evaluation procedure that was described in Section 3.7. The same interview guidelines were used for IP1, IP2, IP3, IP4, IP5, and IP6. They were asked identical questions, enabling a direct comparison and the observation of views from different perspectives. Because IP7 was a traditional real estate marketer rather than a blockchain expert, a customized interview guideline was designed for them. IP7's input features sporadically in the exposition because that interview only yielded a few relevant points. Direct quotations are used to present the results. They have been translated from the transcript, which is in German. In this context, IP1, IP2, IP3, IP4, IP5, and IP6 are treated as blockchain experts; IP7 is treated as an expert in traditional real estate.

This chapter is divided into three sections. The first is about blockchain in general. It overviews the technology and explores opinions about its current state, the areas in which the interviewees see the strongest potential, and areas in which there is room for improvement. The next two sections concern the tokenization of physical real estate and virtual real estate. These sections form the main and most relevant part of the analysis because they are directly related to the research question. The two sections concern challenges as well as opportunities. They serve to generate understanding of the topic and to characterize the two types of real estate. Furthermore, the findings are intended to raise awareness of the status quo and the actual process of tokenization as well as its elements. These findings are used to distinguish between physical and virtual real estate in the context of tokenization. The aim of this chapter is to create a foundation for the answer to the research question that is presented in the last chapter.

4.1 Blockchain in general

Potential areas

The answers given by the experts indicated that they prioritized the financial sector when describing the potential of blockchain technology. However, there were differences in approach and characteristics. IP1, IP2, IP3 and IP4 saw enormous potential in the financial world. IP2 added that before this potential can be harnessed, blockchain would need to develop well in unrelated fields. IP2 thus addressed the entire value chain. They explained that the process will take even longer in finance due to the lack of regulatory standardization. IP5 saw potential in many different use cases and described blockchain technology in general as the infrastructure of the future, which still needed time to develop. IP3 built on this proposition. They saw the greatest potential in the asset area, especially in illiquid assets. Time was still needed to develop this potential properly. IP3 spoke of democratizing markets in a theoretical sense because no-one had really achieved that outcome yet. IP6, conversely, had a clear vision of the areas in which blockchain had the strongest potential:

Personally, I see the greatest potential where you do not have traditional or classic central registers for valuable objects. Especially with real assets [...] where you do not have a register like a land register or a share register or a securities register, but where everything simply changes somehow opaquely from hand to hand [...] because it is a very cost-effective solution to register, transfer, and share.

Criticism on blockchain

The blockchain holds considerable promise, which requires time to develop. There are also aspects that were criticized. The interviewees had very different opinions. IP1 and IP2 did not see any points that were worthy of criticism. IP1 stated that technologies in general should not be criticized, but IP1 emphasized the public debates on energy consumption. However, IP1 made it clear that the problem has to

do with energy use and not with energy consumption: individuals are willing to use energy for the technology and therefore view it less critically. The only questionable issue that IP1 mentioned was that many dubious personalities are involved in blockchain. IP2 said that all of the criticisms that they would normally refer to had already been addressed, and thus they could not criticize the blockchain. However, IP2 saw user friendliness and user experience as issues and explained that losing a private key can result in losing all of one's money. Therefore, it is difficult to imagine that individuals would be willing to manage their own private keys. This is not a technological issue but a problem with end users. IP3 and IP4 agreed that the blockchain is still in its early stages, making it difficult to predict the future. In this context, IP4 went on to explain that it is not the blockchain that should be condemned but expectations about it. The entire concept needs time to develop. IP4 criticized the tendency to handle problems at the expense of security. IP4 used the example of switching to a different blockchain to save transaction costs while ignoring safety concerns. IP5 and IP6 spoke in a coarser tone than the other experts. IP5 characterized blockchain technology as a double-edged sword, criticizing energy consumption but also explaining how all blockchains are becoming more efficient in their use of energy. Furthermore, IP5 brought up an intriguing argument about transparency. Transparency is typically considered as a benefit. However, for IP5, it goes too far. IP5 considered public blockchains to be overly transparent, particularly in the context of the Internet of Things (IoT), where data is transmitted constantly. IP6 identified several flaws in the system. They considered decentralization to be the most significant among them. Decentralization, according to IP6, is always connected with a loss of efficiency, despite the safety and redundancy of a decentralized network. IP6 perceived centralized solutions as superior and more efficient. From this perspective, IP6 cast doubt on the decentralized nature of the whole system. To summarize, different viewpoints were expressed, but the majority of experts were critical of the manner in which blockchain is used and the manner in which individuals interact with it rather than of the technology *per se*. They believed that it was too early to draw conclusions.

4.2 Tokenization of physical real estate

Experts' insights

The second part of the interviews concerned the specific applications of blockchain tokenization in physical real estate. An attempt was made to cover as many perspectives as possible in order to obtain a broad understanding of the subject. Tokenization, awareness, challenges, and opportunities were all analyzed. Regulation was also examined in detail. The development of awareness follows a favorable pattern. According to IP4, the term "Bitcoin" was always used in relation to blockchain three or four years ago, and awareness of what blockchain actually is, was relatively low. However, there has been gradual progress in this area. IP7 substantiated this assertion. As a traditional real estate marketer, their awareness went beyond Bitcoin. They were aware of tokenization and saw much promise in indirect real estate investment, which allows investors to acquire shares in earnings from real estate. However, they found it difficult to envision tokenization in direct transfers of ownership. IP1 and IP6 did not believe that customer or end-user awareness was necessary. IP6 said that a technology should not be at the forefront from the perspective of customers and described blockchain as a means to an end. IP1 took a similar stance, claiming that customers do not need to understand the technology as long as it works. As a result, IP6's company was unsure whether the phrase "tokenization" should be used at all, given that only a few individuals are familiar with it. Another important point that IP6 mentioned was that blockchain has to overcome many prejudices and is typically associated with scams:

“Transparency” is ultimately also a word with the blockchain, but we do not put it at the forefront in our communication with the customer that we use the blockchain, that we tokenize.

As a result, presenting arguments about the benefits of the product was more sensible than elaborating on the technology that makes it feasible.

Turning to the tokenization of real estate, it quickly became apparent that two phrases are mentioned regularly, namely “land register” and “regulation.” Therefore, all of the blockchain experts (IP1, IP2, IP3, IP4, IP5, and IP6) agreed that the advantages and opportunities that are highlighted often in the literature are not yet feasible due to a lack of infrastructure. IP2 elaborated on the main problem of the land register and explained that it was not yet connected to blockchain technology. Therefore, a special purpose vehicle (SPV) was needed. However, IP2 believed that the topic could soon become exciting, provided that the land register co-operates at some point and integrates itself into the blockchain. IP2 saw much added value in business models such as “rent to own,” which offers an individual who lives in a house the opportunity to purchase more ownership every month until they eventually become a full owner. The interview with IP7 showed that the traditional real estate market is functioning and has efficient processes. IP1 said that it is precisely for these reasons that tokenization would not trigger a revolution in that market. However, they hoped that tokenization would improve, modernize the market, and make it more liquid, which would entail numerous benefits. IP3 also made it clear that tokenization can be implemented easily at the technical level by using smart contracts. They also explained that real estate cannot be tokenized directly. IP3 also emphasized the lack of land register infrastructure and the fact that it is legally impossible to tokenize a regulated asset directly. Every jurisdiction has different laws. Therefore, there is no global standard. Unlike the other blockchain experts, IP4 saw tokenization as a revolutionary tool, especially in real estate. This said, many of its features, such as reliance on third parties that store or secure real assets, have nothing to do with the blockchain, which is likely to continue in the future. According to IP4, the potential of the revolution is limited because real estate transfers involve many intricate and manual processes that cannot be mapped onto the blockchain. Tokenization, conversely, has considerable potential that can be realized once its benefits are acknowledged by political bodies, which will abolish the processes in question. IP4 explained the potential of tokenization as follows:

Then, you would have a specific token for your property that represents that the property belongs to you. If you want to sell this property, you can simply transfer the token. If that is set up appropriately, it is as secure as today's very complex process. [...] All these payments-promise issues are completely eliminated; the efficiency gain is incredibly large here, but we still have to wait a few years until we have the legal framework for this.

The enormous bureaucratic effort that individuals who acquire or want to acquire a property must undertake was also addressed by IP5. They explained that an integrated and upgraded land register would eventually allow property to be acquired through a token swap. IP6 compared the token swap mentioned by IP5 with Uniswap and described this specific process in more detail as follows:

[...] owner of an asset, even if you have a fraction of it, can put liquidity, and, in that sense, a certain part of the trading can profit from fees, and that would be absolutely revolutionary, of course, if you could suddenly do that with tokenized real estate or stocks [...]. I am of the opinion that this is not unthinkable and will even come at some point; the question is only when and to what extent do we still need additional laws and securities.

Regulation implementation concerns

Questions about regulations were also asked in this phase of the interviews because the topic surfaces regularly. Since the experts were from German-speaking regions, they were asked about developments in Liechtenstein and Switzerland and the extent to which legislation had aided tokenization. IP1 took a critical approach to the situation, stating that identifying a lawful tokenization business model remains difficult despite the efforts of lawmakers. As a result, IP1 remained skeptical of the process but hoped that the blockchain law in Liechtenstein would lower entry barriers for new business models. However, according to them, whether this will happen remains to be seen. IP3 shared these sentiments and adopted a pragmatic approach to the situation. They claimed that FINMA in Switzerland had been instrumental in attracting attention to Crypto Valley. IP3 saw the definitions of utility tokens and payment tokens and the distinctions between them as helpful and explained that everything else would be treated as an asset or as a security. The benefits of tokenization may not be apparent because many obstacles are yet to be overcome. IP3, speaking on the Blockchain Act in Liechtenstein, said that, like in Switzerland, it adjusted existing laws, which was reasonable. Of course, it was a step in the right direction, but much had to be done to resolve all of the issues. As far as the MiCA draft was concerned, IP3 stated that it had been a pleasant surprise, especially the provisions on secondary markets. IP2 and their company had not been able to derive any direct benefits from the Blockchain Act because they were not involved in tokenization. However, they were delighted to note that Liechtenstein was considered as a pioneering country because of the Blockchain Act and that the European Commission had drawn inspiration from Liechtenstein, as evidenced by the MiCA draft. Speaking on regulation, IP2 also made the following statement:

[...] thanks to the Blockchain Act in Liechtenstein [...] is very optimized and very clean, companies can offer its service. This is why Bittrex, for example, has settled here and has 100,000 customers. Thanks to the Liechtenstein regulatory framework, they have a clean solution here and can operate in Europe without fear.

IP4 and IP6 welcomed the legal developments. According to IP6, the TVTG law in Liechtenstein is a cornerstone that will allow their company to engage in all types of tokenization-related activities as a regulated institution. In addition, IP4 described the regulators in Liechtenstein and Switzerland as highly advanced. IP4 was aware that problems remained, especially with the regulation of DLT trading systems, which allow digital assets that are classified as securities to be traded on exchanges in a regulated manner. Regarding the MiCA draft, IP4 said that political processes meant that change would be much slower than in Switzerland and Liechtenstein. For example, IP4 mentioned the regulation of exchanges on which securities tokens can be traded, which is still impossible in Europe, and mentioned that the passage of suitable regulation would take several years. In comparison, Switzerland and Liechtenstein had already gained experience in these regulatory areas. Like IP2, IP4 described those countries as pioneers.

Challenges of tokenization

Beyond regulation, the interviewees were asked about other challenges in the tokenization of physical real estate. The lack of land register infrastructure was mentioned again by most of the experts, this time at greater length. IP2 made it clear that everything ultimately depends on the land register and shared the following observations:

[...] the land register must know who is behind it. They have to know which person is behind which address. So, the whole know-your-customer process has to be fully integrated and automated, and the land register would have to have a register. [...] the land register is clearly a sticking point because the

land register is not there yet technologically and wants to know who is behind it [...] and because it also earns money with every change of ownership [...].

Additionally, IP6, speaking on the main challenges, mentioned that traditional land registers that are not based on blockchain. Without this infrastructure, it is not possible to tokenize properties directly. IP6 even went as far as to say that without the possibility of transferring ownership, one must return to the topic of tokenizing financial instruments. IP6 mentioned that there are business models under which conventional bonds are tokenized and sold. However, a functioning central market for such products already exists, and IP6 saw no way in which tokenization could add value in this context. IP4 also saw important problem with the transfer of ownership because of the land register but pointed to the current alternatives, which entail further challenges. One alternative solution involves the interposition of a company. IP4 described the resultant challenges as follows:

That can be an SPV that simply holds the real estate. You then tokenize the equity or the borrowed capital of this company. This creates new challenges. On the one hand, on the regulatory side, if you tokenize equity that invests in real estate, maybe even several, you are very close to the fund area, so it may be that you are classified as a fund and not as equity. There are also challenges in the borrowed capital area because you want to distribute, for example, real estate returns that can fluctuate. Borrowed capital usually has fixed interest rates, so the vessels are still not quite suitable in some cases, or it simply has to be discussed in detail with the regulator as to how it can be classified.

IP4 also mentioned two intriguing problems. First, while the technical side of tokenization is not a challenge, mistakes are frequently made, in that a technical token may not legally represent what was promised. Another topic that emerged had to do with business operations. The sale of tokens is frequently underestimated, particularly in smaller projects. Because IP3 had cofounded a platform for real estate tokenization, they possessed intimate knowledge of the issues. The most difficult hurdles, according to them, are the land register and the secondary market. Speaking about the land register, they said:

[...] in the land register, you have natural and legal persons, who may hold or have land ownership. And we specifically propose [...] that a technical person can also be the owner of a plot of land, and a technical person can be a smart contract. [...] in principle, that would be an important step that you can talk about direct tokenization at all, and also to be able to use the essential theoretical advantages of a blockchain.

As far as the secondary market is concerned, IP3 had some direct experiences, on which they made the following observations:

[...] on the secondary market, we also started a company back then, quite simply [a] security token exchange, and in parallel we exchanged very closely with FINMA, and are still working on it. This means that assets can and are allowed be traded in a decentralized manner, which is extremely important and a key factor. Regardless of the area, including real estate, if you can do that, it suddenly becomes interesting. If you can get the secondary market to work in a way that people need it, sellers and buyers or investors and asset owners, then it gets really exciting, but also in terms of liquidity.

IP1 took a different approach and addressed operational histories that were related to the regulatory challenges. For example, what happens if a user loses a wallet that contains tokens that represent a particular property. Theory and textbooks show that decentralized technology eliminates centralized entities. However, in many models, there is still an intermediary that regulates such matters, and tokens can be produced again by that third party. From a regulatory point of view, they can also proscribe certain customers. This institutional setup is also desirable for the end user. Otherwise, if the private key is lost, they would have no recourse against the property. In this context, IP1 summarized the current state of affairs as follows:

They want a backup solution; they want a host, somehow. That means you are always somewhere in between [...] I think it will stay that way. [...] You have to offer an attractive product, attractive service with the right services, but, at the same time, it should not be the case that we are just another intermediary. Basically, we are competing, and many who tokenize are competing to eliminate intermediaries, to make the world simpler. That is why you cannot end up taking the whole old world onto the blockchain and then having an additional intermediary. There are a lot of issues there that we are struggling with.

Potential of tokenization

The final question on the tokenization of physical real estate concerned opportunities in the sector. IP5 answered briefly, saying that tokenization creates a new financial product that simplifies the handling and trading of real estate. Thus, tokenization definitely has potential and could become established in the coming years. IP1 picked up where IP5 had stopped when outlining challenges, focusing on the elimination of intermediaries. In addition, IP1 emphasized that many things remained the same. For example, when a property is maintained, it still needs management. However, if a property is bought and tokens issued, ordinary individuals can enter the real estate market with small amounts of money. Furthermore, IP1 mentioned funds. However, there are many intermediaries, such as banks, issuers, and so on, which accept funds from multiple parties. This tendency leads to many charges in the form of fees. To differentiate itself from funds and to establish its financial attractiveness, the company at which IP1 was a CTO had eliminated numerous middlemen. IP1 described the benefits of this model as follows:

[...] as soon as the customer invests, brings in money, then their money is tokenized, [...] then we are completely in the blockchain ecosystem. [...] because we can do blockchain transactions very cheaply, we can do interest pay-outs. For example, we do monthly interest pay-outs. Typically, tenants pay interest monthly, and we can pay that out to the investor right away. That is relatively uncommon, and we can do that relatively efficiently because everything is tokenized. Or, even if the customer wants to restructure his portfolio, he can say [...] "I want to take away 100 tokens from Property A and move them to Property B," which does not incur any fees for them or for us.

When asked about opportunities, IP2 and IP3 also addressed the elimination of intermediaries. IP2 saw considerable opportunities, not necessarily because of possibilities for fractional ownership but because transfers would be much easier to complete and would not require the involvement of multiple parties. In addition, there would be no need to visit the land register to change the names on deeds because all the necessary operations would be executed immediately when the token is transferred. IP2 focused on security and the reduction of work processes. IP6 highlighted fractional ownership and envisioned a variety of use cases for it. Fractional ownership structures have the potential to improve the overall efficiency of the market. For instance, IP6 mentioned inheritance, which can be handled more rapidly and transparently when ownership must be split. Unlike traditional inheritance procedures, which require significant administrative effort, the division of property could be accomplished easily through tokenization. Moreover, IP3 noted that anyone could participate in the market for tokenized assets and referred to models that entail less reliance on banks. They made the following comment:

Simply put, if I need money or a mortgage, instead of going to a bank, I can also use certain platforms or protocols that already exist [...] on Ethereum. So, instead of a cryptocurrency, I can offer certain tokens of my own property as collateral, and I can take out a loan in a matter of minutes.

IP3 also mentioned the rental business, which is often underestimated. New business models, such as "rent to own," could emerge in this environment, as mentioned by IP2 at the beginning of the interview. IP3 added that such models could develop due to lower transaction costs, simplified traceability, and the

legal certainty that the blockchain produces. IP4 emphasized the efficiency of transfers and the democratization of asset classes, which would allow those who could not previously invest in real estate to do so by using a token. Furthermore, IP4 referred to funds, like IP1, but stressed the benefits of tokenization from a different perspective. According to IP4, when an individual buys a share in a fund, they do not know what is behind it. Furthermore, they cannot influence its composition. Tokenization allows users to choose between real estate assets directly and to build individual portfolios, resulting in improved flexibility and more power.

4.3 Tokenization of virtual real estate

The results of the last part of the interview, which covered tokenization in the context of virtual real estate, are presented in this section. Since the phenomenon under observation is relatively new and the literature contains no predictions about its future, the goal was to determine what value these developments could add to the market and to isolate the characteristics of virtual real estate so that a comparison can be made between physical and virtual real estate in terms of tokenization.

Relevance of buying NFTs and its added value

Before elaborating on motivations and added value, IP5 said that there are a few different ways to obtain virtual real estate in the form of an NFT. It is possible to mint virtual real estate or to buy it on the secondary market, making the operation relatively simple—it only requires a wallet. Furthermore, IP5, along with IP2, IP6, and IP7, spoke on the speculative nature of these initiatives, emphasizing that only the most ardent supporters believe in them truly and that most investors are interested exclusively in profiting from virtual land. However, IP5 was optimistic that certain initiatives would succeed, especially in the area of video game assets. IP3 mentioned the gaming industry, which is not to be underestimated because it attracts considerable investment. Beyond hype and gaming, IP3 did not see virtual real estate as particularly promising. Nevertheless, IP3 had an ingenious idea about NFTs. They gave the following explanation:

[...] there are certain absolutely super benefits that you can program. As an example, within a smart contract, the original artist or creator of anything will still be paid or rewarded in the future.

Additionally, IP3 made an exciting point, namely that there were similar stories before blockchain, such as Second Life. This example was developed further by IP1. Second Life did not rely on blockchain and offered virtual lives, virtual offices, and much more, along with gamification elements. It worked. This phenomenon was also addressed by IP2, who made it clear that Second Life worked with a central database. They saw no reason why virtual real estate or similar applications would perform better in a decentralized environment. IP1 did not see much utility in most NFT projects and thought that they would eventually be worthless. They compared NFTs to collectibles and argued that the development of NFTs did not matter, provided that it remained entertaining. They considered NFTs a gimmick. Of course, the identity of the party that comes to dominate the market also matters, but IP1 did not see any benefit in Decentraland, the most advanced project. IP2 added that development or enforcement would ultimately determine who would prevail in the competition between centralized projects, such as Meta (Facebook), and decentralized ones, such as Decentraland. Nevertheless, this topic is interesting from the point of view of the company that IP2 serves as CEO due to the enormous increase in value that is associated with it. IP2 explained that they purchased virtual land in Decentraland early and rented it to third parties, greatly enhancing its value. Again, much hype surrounds the topic, and it has been addressed multiple times. According to IP2, whether the entire phenomenon should be based on centralized

or decentralized servers is unclear. For the end user, it is also irrelevant. In response to the topic of hype, IP2 gave the example of Snoop Dogg purchasing virtual real estate, causing the price of plots around his to skyrocket. The user can roam the world of a player and buy properties in intriguing locations. IP7 drew an analogy to the conventional real estate market and showed that virtual real estate is purchased for image purposes, such as reputation or conspicuous consumption, which also occurs in the traditional market. Properties in prime locations are far more expensive in the traditional market, and this trend is replicated in the virtual world. IP2 saw considerable potential for corporations such as Adidas because they can develop apparel and collections for the metaverse and earn large profits. Unlike end users, added value is manageable. There are potential use cases, such as viewing and assembling 3D automobiles in digital worlds. Whether this requires the use of blockchain technology is debatable. IP2 provided the following explanation:

I am convinced that in this NFT topic, regarding metaverse and gamification happenings will take place more in centralized structure because the providers know the added value but do not want to hand over the free will to the users. For example, Meta, respectively Facebook [...] will charge 47% of all minting fees, so if I sell something for CHF 100, I will pay CHF 47 to Meta. [...] that is a gigantically big branch of business.”

IP4 adopted a more optimistic perspective, arguing that added value would emerge in various ways over time. According to them, the basic prerequisite was a belief in the metaverse. According to IP4, individuals would not spend more time on the internet due to the emergence of the metaverse but would instead adapt their current behavior and optimize it. IP4 described the following scenario:

Instead of having our 20 tabs open and comparing hotels, I was in the metaverse and briefly looked at the hotel in Turkey. Or, we went to Greece briefly and did some wandering around and decided on that [...]. Then, of course, the questions are, if we are physically there on the internet, where do we walk through there? One possibility is that you walk through cities there, there is life, there is real estate, and then it makes perfect sense to own this real estate because it could have an increase in value.

Comparison between physical and virtual real estate

Finally, the last part of the interview was also intended to explore the differences between virtual real estate and physical real estate in terms of tokenization and the distinction between the two types of assets. IP7 saw virtual real estate as being uninhabitable, unlike the physical version, and focused on the particular need for housing, which took on a new meaning during the pandemic. Thus, the two markets are completely different. IP5 corroborated IP7's statement. They considered digital land to be useless *at present* because society is not yet at a point at which it can make use of it. IP2 also emphasized housing:

[...] physical real estate, [...] I can look at, I can build a house on it, I can live in it. In the metaverse, I cannot live. So, if I live in a 20 m² apartment, but in the metaverse I have a 5,000 m² mansion... but in the end, it does not bring me any value. [...] the digital world is a pure show off to the outside, and some show for certain people. Physical real estate is a hard asset, a rare good of which there is no more. Yes, logically, in a digital world you can also limit that, but digital real estate makes too little sense for me personally [...].

IP4 and IP6 compared these phenomena in legal terms and classified them as being fundamentally different. The two categories serve different purposes. Furthermore, IP4 identified the development of employment profiles that are similar but differ in terms of activity as intriguing. IP4 used the example of a real estate agent. In a virtual environment, they would be required to develop a different set of technical skills and would be asked to focus on a completely different target group. Even in architecture, the design of virtual properties requires a different technical skillset than the design of physical ones. Since

anything can happen in a digital environment, IP3 defined virtual real estate as being much simpler. Therefore, the tokenization of virtual real estate, from a legal perspective, simply refers to digital objects. In comparison, tokenizing physical real estate is far more difficult because a seamless connection must be established between the physical and digital worlds. This problem is considerably more complex than instances in which all relevant activity occurs in a digital environment. However, if these issues are resolved soon, IP3 believes that the tokenization of physical real estate would become very promising. IP3 also mentioned that the process may no longer be called “tokenization” by then and that it may be designed differently. From a technical standpoint, IP1 described the tokenization of virtual real estate as a simple process that is nearly identical to the tokenization of physical real estate, which can be mapped with the same token standard. From an investment point of view, the two are completely different. Furthermore, IP1 explained that virtual real estate will remain playful for the foreseeable future because it is connected to gaming. That does not mean that it cannot generate billions. However, IP1 saw no real points of contact with the real estate industry:

I cannot think of any agency at the moment that would offer you an apartment in Zurich or in Liechtenstein and in Decentraland.

5 Discussion

In this master's thesis, empirical research was conducted on the tokenization of physical and virtual real estate. Seven experts who were described in detail in chapter 3.6 were consulted after guided interviews on this subject were conducted.

The empirical results from chapter 4 serve as a basis for answering the research question and are discussed in the following chapter. These findings, when combined with the theoretical underpinnings presented, allow for the derivation of consequences for research and practice. Through this approach, it should be possible to make clear statements that serve to raise awareness and highlight the potential and challenges of tokenization in the areas of physical and virtual real estate. Furthermore, a comparison of these two variables will be conducted to identify differences. Visualizations and frameworks will be developed to facilitate comprehension and simplicity.

5.1 Blockchain as underlying technology of tokenization

Physical and virtual real estate are being tokenized as a result of blockchain technology, which also makes tokenization practical in this context. As one of the objectives of this master's thesis is focused on awareness, blockchain technology has been presented in detail along with the most important elements and features needed to understand tokenization. Blockchain technology has to fight many prejudices, because cryptocurrencies like Bitcoin or Ethereum are often misused for criminal activities (Kethineni & Cao, 2020, p. 325). This was also confirmed by IP6, who stated that initiatives are frequently labelled as frauds without first giving them a deeper look since blockchain often needs to deal with preconceptions. But this criticism of the blockchain is slowly moving more and more into the background. The reason for this is that the blockchain has much more to offer than just serving as a digital currency and shows enormous potential in various industries. This development could also be seen in the interviews as only one of the experts raised the issue of misuse of blockchain by scam projects. The reason for this is that they are intensively engaged with this technology and recognize the benefits, despite the fact that, as IP4 explained, blockchain is still in its infancy. It is still unknown in which direction the entire concept is developing. Thus, the majority of the experts were critical of the technology itself, such as energy consumption. This is a good key point to discuss this important aspect and to refer to chapter 2.1.4 where the consensus algorithm and the two most commonly used algorithms, proof of work and proof of stake, are addressed and explained. Both consensus approaches have different advantages and disadvantages. For instance, proof of work is frequently criticized since it consumes a lot of energy. Such developments must be taken into account since new consensus algorithms are continually improved, and as IP4 stated, this requires that the entire blockchain concept be updated for it to function. Because it frequently happens, as IP4 explained, those problems are now solved by just moving to an unreliable blockchain.

However, the potential of the technology outweighs the technology, which is why, as stated by IP1, we should not speak of energy wasting or consumption but rather of energy usage, since people utilize energy for this technology to create new innovations. Blockchain experts are quite optimistic about the prospects, particularly for the financial sector.

The author, Baum (2021), makes it clearly evident that blockchain technology familiarity and awareness are prerequisites for market players in tokenization (Baum, 2021, p. 41). This can only be partially confirmed based on this empirical research. The awareness of blockchain technology has changed according to IP4 and sees positive progress in this respect. The existing research can therefore be refuted

or rather supplemented in this case, because IP1 and IP6 made it quite clear that, unlike the providers who use this technology for their services or products, the end consumers, or users respectively do not require any awareness. The reason is quite straightforward as IP6 explained, as long as everything works, the customer does not need to understand the technology if the outcomes are what they are expected. As a result, the technology should fade into the background. This can be compared to virtual real estate; as long as the end user can move around in the virtual world without any problems and can buy and sell virtual real estate without any problems, the technology does not play a major role.

5.2 Tokenization of physical real estate

This chapter provides a comprehensive basic understanding of tokenization of physical real estate and virtual real estate, in conjunction with the next chapter. These two chapters provide the basic framework for comparing the two and answering the research question at its core.

Chapter 2.2.1 presents real-world cases and shows how the tokenization of physical real estate is currently being handled. When examining these practical examples, direct ownership is not yet attainable, as evidenced by the real-world cases described. Several ICO projects, such as Crowdlitoken, provide investors the opportunity to become owners of an asset for a minimal investment, according to the authors Garcia Teruel and Simón-Moreno (2021). When examining the specific conditions and the smart contract codes, it became evident that one company commonly owns the tokenized asset, while another manages it and makes the necessary decisions regarding its use and disposal (Garcia-Teruel & Simón-Moreno, 2021, p. 3). Therefore, as shown in Figure 4, direct tokenization of physical real estate and direct ownership transfer are not yet possible to achieve. Only profit participation rights or bonds can currently get tokenized by tokenization projects. Real estate can be transferred with economic rights but not ownership in a variety of methods. Tokenized equity can be used as a share in real estate investments. As confirmed by IP4 and IP2 statements on the subject, this appears to be true for both the tokenized world of indirect real estate investments. IP4 mentioned that equity or the borrowed capital of the company can be tokenized by the support of a Special Purpose Vehicle (SPV). IP2 added that a Special Purpose Vehicle (SPV) is needed as the land registry is not connected yet to blockchain technology. For instance, a real estate business may purchase the property, and investors could engage in the firm as equity investors through shares or participation certificates or as debt investors through bonds.

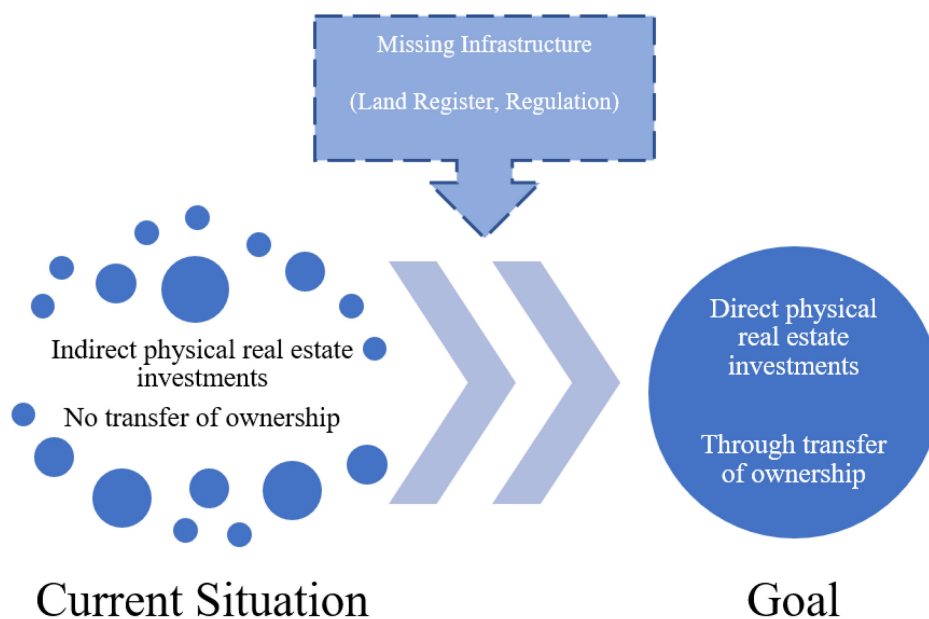


Figure 4: Status quo of physical real estate in terms of tokenization

Source: Own illustration

This is due to the fact that tokenization as a blockchain application is still in its infancy and as such faces numerous hurdles, most of which are legal and technological in nature (G. Wang & Nixon, 2021, p. 3). These legal and technological issues are also covered by the authors Heines et al. (2021), who discuss tokenization in general without mentioning real estate. However, the primary focus is on the regulatory challenge. Furthermore, recent research on tokenization has a strong emphasis on the financial industry's potential and challenges (Heines et al., 2021, p. 4.).

A similar initial situation can also be observed in the discussion of the tokenization of physical real estate. However, with regard to real estate, the literature has almost exclusively reported on the potential and benefits, while the challenges have been relegated to the background. The potential frequently discussed in the literature has been validated by all six blockchain experts, despite the fact that they are not yet applicable in today's world. The expert interviews, however, can disprove the technological difficulties as a challenge. IP1 describes the tokenization process as generally very simple, regardless of what is tokenized. This can also be confirmed by IP4, as IP4 characterized the technical side of tokenization as not challenging at all but emphasizes that mistakes are made in that the technical token does not legally represent what should be represented. When the insights into the tokenization process were inquired about as an introduction, it was clear that challenges play a significant part in the tokenization of physical real estate. In the first portion of the general interview, the terms "regulation" and "land registration" were referenced multiple times. The author discovered the significant challenge of the land registry only through the interviews, which could not be identified by the intensive examination of the literature, which explicitly dealt with the tokenization of physical real estate in the literature review.

In terms of their land registry systems, countries can currently be divided into three categories. The countries in category 1 have a highly well-organized land registry. These countries, which include the United States, the United Kingdom, Holland, and Sweden, are attempting to enhance the quality of their land registries through cutting-edge technology. Countries in category 2 have less efficient land registries. These nations' bureaucracies, antiquated processes and systems, complicated procedures, and even corruption and fraud are the causes of this. Greece is used to illustrate this category. The third and final

group includes countries that operate without or ineffectively handling land registration. Similar issues to those in category 2 are present in this category but to a greater degree. In Ghana, for example, 80 percent of landowners do not have title deeds. As a result, most land registers are constrained in various ways. On the one hand, it is hardly practical to introduce new business models, as established practices impede this. Furthermore, many of the current land registry operations are too expensive and time-consuming, allowing weeks-long processes to be completed in a matter of hours or minutes. An automated land registry system could eliminate the existing problem of too many middlemen in real estate transactions. Additionally, there are security concerns and a high level of mistake tolerance. Finally, there are issues with fraud and corruption that primarily affect categories 2 and 3 (Themistocelous, 2018, pp. 195-197). These issues, and many more, could be resolved and raised to a new level by the adoption of blockchain in the land registry (Themistocelous, 2018, pp. 199).

The interview made it apparent that the land registration must run on the blockchain. Currently, this is not the case; as a result, a business model like direct fractional ownership of a property is not currently practical due to a lack of infrastructure through the land registry. IP2 made it clear in this context that the tokenization of physical real estate is only really feasible if the land registry plays along. Therefore, IP6 also went so far as to say that, until there is no blockchain-based land registry but rather a traditional one, IP6 cannot see any real added value for tokenization in the commercial real estate sector at the present time. To summarize the challenge on the land registry, no direct tokenization can occur as long as the land registry is not on the blockchain.

The previously mentioned regulatory component is another one of the key issues that emerged very clearly from the expert interviews. IP4 makes it clear that a global standard does not exist and, therefore, each country has its own rules, which makes direct tokenization impossible. And it is estimated that it will take years before such a legal framework is established.

The definitions alone present a challenge for regulation. As will be demonstrated in section 2.1.6, there are no universal or established definitions for tokens. Payment tokens, utility tokens, and security tokens are distinguished by FINMA in Switzerland and BaFin in Germany (FINMA, 2018, p.2; BaFin, 2019, pp. 5-6). Tokens have yet to be put into a universally recognized or international classification. As a result, such categorizations of tokens serve primarily as aids to understanding (Markheim & Berentsen, 2021, p. 63). Although categories and definitions can be extracted from the literature, according to writers Wang and Nixon (2021), these are not universally accepted (Wang & Nixon, 2021, p. 3-4). These flaws in various definitions, which can be found in many forms in many countries, make the standardized cross-national legislative process more difficult. Regulatory authorities in several nations are attempting to solve this through government and industry regulations, which are hardly in place yet. No country has well-developed regulations for such transactions yet, despite the fact that regulators in several nations are starting to consider rules for trading tokens and cryptocurrencies. Problematic legal issues also need to be clarified. For example, what happens if the business holding the underlying real asset sells the land? What happens if the asset is stolen or harmed (Sazandrishvili, 2020, p. 5)? According to IP1, another question is what happens if a user misplaces his wallet that contains tokens representing a particular property, emphasizing the importance of regulation through several levels. However, legislators have addressed this issue and are working to fill these gaps and provide a framework in the process. Chapter 2.1.8 shows how the countries of Switzerland and Liechtenstein are currently handling this process and are also considered pioneers in the regulatory field. There is no cross-border regulation yet, but the European Commission is trying to create a cross-border legal regulatory framework through the MiCA draft (Bočánek, 2021, p. 38). This development was also followed by blockchain experts. The regulatory aspect was viewed very critically but was described as a step in the right direction. After all,

the experts are aware of the significant challenges faced by the regulations and that it will probably take several years before it is possible to transfer ownership directly through tokenization. However, countries such as Switzerland and Liechtenstein have already gained experience in this area of regulation, as IP4 clarified. Furthermore, the European Commission, which administers cross-border regulation, will take considerably more time. Nevertheless, these first regulatory adjustments in Switzerland and Liechtenstein have focused on modifying existing laws, rather than creating new ones.

The authors, Tien et al. (2019), also state that international regulations are not in a position to provide a corresponding regulatory framework in the near future. The authors believe that the aforementioned modification of existing laws is important, as overregulation by too many new laws would undermine the benefits of tokenization and thus also limit the innovation process of companies. On the other hand, a lack of regulation makes prospective participants hesitant to use the technology. Furthermore, fraud and hacks on unregulated security tokens exchanges will impede the security market's long-term growth and maturity (Tian et al., 2019, pp. 7-8).

Due to these existing challenges, the advantages of tokenization of physical real estate, which are often discussed in the literature, should only be taken with caution and therefore remain theoretical, as there are deficiencies in the existing regulations and regulatory uncertainty, as already mentioned (Markheim & Berentsen, 2021, p. 59). However, if these challenges are overcome in the coming years, as predicted by the blockchain experts, the potential presented in detail in chapter 2.2.2, which include benefits such as global access, liquidity, removal of intermediaries, faster transactions, transparency and immutability, which literature regularly addresses, can be achieved.

The blockchain experts agree with the benefits outlined in the literature with a few changes and anticipate enormous potential once the obstacles related to land registry and regulation have been removed. The elimination of intermediaries is frequently mentioned and regarded as one of the main potentials, in addition to the easier accessibility and entry into the real estate market, even for investors with less money, and the prospect of shared ownership. But here it must be emphasized, as IP1 clarified, that not all central intermediaries will be eliminated, as often described in the literature. From the point of view of customers and end users, a third party is which, for example, regulates whether a lost private key can be restored and offers the users a certain level of protection. Thus, customers want a back-up solution that makes the complete elimination of central institutions impossible. Nevertheless, it goes without saying that the majority of middlemen should be eliminated, therefore the blockchain shouldn't merely be added to the existing world. Further innovative potential, which builds on the existing fundamental benefit, would be possible based on the benefits described, such as fractional ownership, as explained by IP6. For instance, distributing illiquid assets more simply in the event of inheritance would be much more efficient and transparent. But tokenization can also establish potential in the rental sector, which is often pushed into the background in connection with tokenization. Completely new business models can emerge from this. IP2 and IP3 mentioned rent-to-own as an example of an innovative business model that allows tenants to buy more and more parts of a property or apartment, which could result in full ownership. The rent would also decrease accordingly in percentage terms if more shares of a certain physical real estate asset were purchased. People often also question why tokenization adds value in terms of democratizing asset classes. Funds are often cited as an already functioning alternative and argument, but IP4 made clear distinctions here and demonstrated the benefits and potential. When it comes to funds, the customer has no exact knowledge of what is in a fund and what goes into a fund. Tokenization offers a more transparent solution in this case, allowing the customer to decide which specific properties to invest in. Thus, it can be stated that the tokenization of physical real estate faces enormous hurdles, especially when it comes to directly transferring ownership. The main challenges are

related to non-blockchain-based land registers and regulatory challenges. After overcoming these challenges, which will probably take several years, tokenization in physical real estate offers great advantages and potential, which are far from being fully known due to its early stage.

5.3 Tokenization virtual real estate

Chapter 2.3, together with its subchapters, describes what virtual real estate is about in the context of tokenization. Thus, the term “virtual real estate” is used to describe the purchase of land or property in a virtual world. The phrase "NFT" is frequently used in conjunction with the phrase "virtual real estate." This is due to the fact that virtual property is traded as an NFT in virtual worlds such as in the Decentraland universe. The customer can use the purchased land in a variety of ways; there are no restrictions and an unlimited range of choices are available; the only limitation is the buyer's imagination (Dowling, 2021, p. 1). NFTs, as described in chapter 2.3.1, first experienced a lot of hype in the art industry, but this hype has taken a new direction due to the emergence of virtual real estate.

It is similarly hard to avoid using the word "metaverse" while discussing NFTs and virtual real estate. Although the word "metaverse" has been used for many decades, this phenomenon has just recently attracted new level of attention. One of the factors behind this development is the CEO of Facebook, Marc Zuckerberg, who has renamed his company to META and set the metaverse as his new goal. This has set the ball rolling, triggering a hype shift into virtual worlds along with virtual real estate in the form of NFTs (Narin, 2021, p. 17). However, it is difficult to identify a pattern to determine the direction the metaverse, including virtual real estate, is moving. A proper definition of a metaverse is already difficult to come by, but experts and researchers agree on a variety of essential characteristics a metaverse must have in order to be considered one at all (Kiong, 2022, pp. 22-24).

There is not much information in the literature regarding the potential and challenges of the metaverse and the associated virtual real estate because, as was already mentioned, this phenomenon is still in its infancy, and it is even uncertain how it will develop. NFTs, which stand in for virtual real estate, do, however, create certain challenges, as demonstrated in Chapter 2.3.3. According to Wang et al. (2021), these challenges can be classified into four main categories: usability challenges, security and privacy issues, governance considerations and extensibility issues (Q. Wang et al., 2021, pp. 13-15). Because it is much less complex, based on the observation of recent years, the literature already provides fewer hints as to how the metaverse might add value. Researchers believe that gaming is the most popular application of the metaverse, followed by social networking and commercial applications (Kiong, 2022, pp. 31-33).

The outcomes of the expert interviews generally support the research findings in the literature. Through a wallet, it is relatively simple to purchase virtual property in the form of an NFT. Naturally, the experts pointed out the initial speculative intent, and as a result, they also claimed that this development adds no value beyond buying a virtual plot of land and reselling it for more. The results showed, however, that it is simply too early and that perhaps people are not yet ready to recognize the true value of this development. Numerous projects are currently active in the metaverse sector. Popular examples are Decentraland, a decentralized project, and Meta (Facebook), a central project. The amount to which the metaverse finally grows and how it will develop, therefore, depends on considerations like which big player will prevail.

Nevertheless, the experts in the interview also showed that the metaverse, which is characterized by gamification elements, has benefits in the gaming sector. IP5 made it clear that potential through NFTs

is evident in the gaming industry. IP3 also believes that the gaming industry should not be underestimated and can recognize the potential in this area. It is not relevant to the end user whether this will occur in a decentralized or a centralized world, and thus IP1 once again addresses the theme that technology should not be the focus. However, the metaverse has a great deal of promise, especially for companies like the one mentioned in the interview — Adidas. Considering that they may own virtual properties, sell their items and collections in a digital environment, and make astronomical profits, the metaverse not only offers opportunities and potential for suppliers but also for customers. For instance, a car manufacturer could open a store in such a virtual space by purchasing virtual real estate in the form of a NFT and providing its services, allowing customers to customize and design their ideal vehicle themselves. Such a 3D setting is ideal for such use cases. Thus, it can be concluded that there are several justifications for purchasing virtual property. Buying or trading virtual real estate can be done for playful, collectible, or bragging reasons. Speculation is undoubtedly one of the key motives for purchasing virtual property at the moment. Although the technology is not yet fully developed and experts and researchers disagree on how the metaverse will develop in the coming years and, as a result, the relevance of virtual real estate, it is still feasible to identify additional value from both the provider and end-user viewpoints.

5.4 Differences of physical and virtual real estate in terms of tokenization

The following chapter answers the core of the research question that deals with the difference between physical and virtual real estate in relation to tokenization. At the end of this chapter, a summarizing framework will visualize the differences for comprehensibility.

The following Figure 5 represents the action outlines of the literature and combines it with the findings of the expert interviews. The conceptual framework connects the important variables and concepts in this study to obtain the target objectives and answer the research problem. This summarizes and simplifies the whole dataset results.

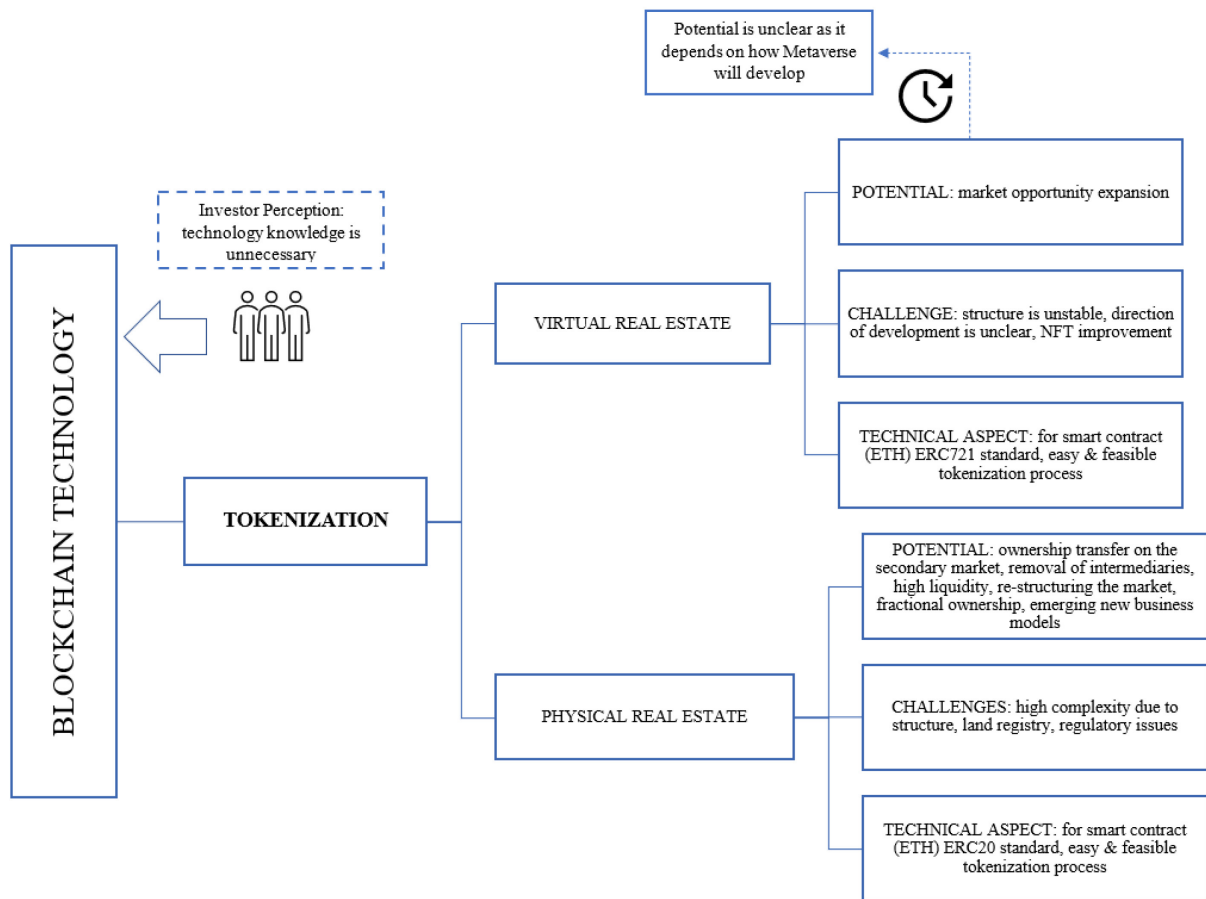


Figure 5: Differentiation of physical real estate and virtual real estate in application of tokenization

Source: Own illustration

Although tokenization of physical real estate and virtual real estate both involve the term "real estate", they are fundamentally different in nature and yet have certain similarities in terms of various aspects. Chapters 2.2 and 2.3 show that most tokens, in terms of physical and virtual real estate, are created on the Ethereum blockchain (Markheim & Berentsen, 2021, p. 68; Baytaş et al., 2022, p. 1). On a technical level, the ERC20 standard is used for physical real estate and the ERC721 standard for virtual real estate. This also becomes apparent when looking at the real-world application examples in chapter 2.2.1. Crowdlitoken and Blockimmo both use the ERC20 standard for the tokens they issue. In contrast, the metaverse project Decentraland, described in chapter 2.3.1, uses the ERC721 token standard for virtual real estate (Kräussl & Tugnetti, 2022, p. 5). Furthermore, both are also very easy to create from a technical point of view. According to IP1, it is a very simple process, and it makes little difference whether a physical real estate or a virtual real estate is tokenized. It is even possible to map both to the same token standard.

When looking at the challenges, tokenization of physical real estate has far greater hurdles for the simple reason that it is trying to build a bridge between the physical and digital worlds and thus create a digital twin. Virtual real estate, on the other hand, as IP3 reported, is in a digital environment from A to Z, which makes virtual real estate in general much easier. Thus, both types are also legally classified completely differently. This is also confirmed by IP4 and IP6. By comparison, it can be concluded that since virtual real estate is much less complex, it is also much simpler to control from a legal perspective. Moreover, the challenges of both types show that the issues of tokenizing physical real estate depend on achieving the desired idea of direct tokenization of a property. The challenges facing the tokenization of virtual real estate are based on improving its existing concept. This can also be confirmed by Popescu

(2021). The author compares the concept of tokenizing actual real estate with that of Decentraland, a virtual environment. By allowing for the purchase of NFTs that stand in for usable virtual real estate, real estate has reached a new level. However, tokenization of physical real estate has not yet reached the development stage compared to virtual real estate, despite the fact that the concept of tokenizing real estate has existed longer than the idea of virtual real estate. The reason for this is due to the fact that real-world deed structures are quite complicated and that many different entities act as middlemen for ownership and validation (Popescu, 2021, p. 28).

Even when discussing potential, there are substantial differences because of the characteristics of these assets. The primary difference between the potentials is that, as can be seen from the discussion in chapters 5.2 and 5.3, in the case of physical real estate, the potentials are already well-defined in the theory and literature, but there are main challenges to be overcome, such as regulatory and land register insufficiencies, because the listed potentials are considered for a direct tokenization of property. In contrast, the potential for virtual real estate is still a bit more unclear, although it has fewer hurdles to overcome. The main issue in this area is that no one knows which direction the metaverse will take or how important virtual real estate will be in this digital environment. This can also be confirmed by the literature. The ambiguity of the metaverse's future is emphasized by the fact that researchers are already investigating where the metaverse is leading us and trying to discover solutions to many problems, some of which cannot presently be addressed (Cui et al., 2022, p. 1).

	Physical Real estate	Virtual Real Estate
Purchase Reason	Housing, basic needs, reputation	Speculative, entertainment, reputation, commercial purpose
Rarity	Rare asset due to geographical constraint	Rare asset due to limited virtual plots of land
Physical Laws	Exist	Non-existent
Pricing	Price effect due to location attractiveness impact (e.g., higher prices at city)	

Table 3: Characteristics of physical real estate and virtual real estate

Source: Own illustration

The Table 3 above shows distinct characteristics of physical real estate and virtual real estate. According to purchase rationale, rarity, physical laws, and pricing. Because of their inherent differences, the two types of real estate are, in fact, whole different markets. The housing element is thus one of the primary differences. Humans can live in homes on physical real estate, which provides a distinctive value and a fundamental requirement for people. As a result, eating, sleeping and other activities that take place in the physical properties. In contrast, virtual properties are commercial in nature and therefore relate to retail, leisure, and social activities, and therefore do not cover basic needs. Furthermore, the professions connected to physical and virtual real estate also differ in other ways. As noted by IP4, professions like real estate agents would have quite distinct skills in a digital environment. Because of how different this environment is from the actual world when it comes to building development, architects must also be technically adept in it.

On the other hand, there are similarities between physical and virtual real estate that can be identified. As IP7 described the traditional real estate market and physical real estate itself, real estate in the real

world is often purchased for image reasons. Especially when it comes to choosing where the location of the real estate should be. Real estate thus costs more, for example, when it is located in a central city, is in a more desirable location, or is inhabited by upper-class residents. As an illustration, IP2 stated that this development is also noticeable in virtual real estate, where the prices of nearby properties increased because of Snoop Dogg's purchase of a virtual property. And for this reason, people purchase virtual estate in the metaverse not only for fun purposes but also to improve their reputation or image.

The authors, Goldberg et al., also attested to these similarities. They discovered through their empirical investigations of price creation that, just like in real life, desirable locations of virtual real estate significantly increase the price. As a result, virtual real estate is rather expensive if it is situated close to a city centre, major roadways, or a business sector, and investors are willing to pay this price increase. But in this context, more variance can be found (Goldberg et al., 2022, pp. 31-32). For instance, since there are no transportation expenses in the metaverse, commuting traffic is less significant than it is in the actual world. Due to the possibility of random encounters, investors acquire real estate in the physical world. However, because teleportation in the metaverse is instantaneous, this impact is not there (Goldberg et al., 2022, p. 29). In a similar manner, it can be deduced that the required infrastructure for virtual real estate, such as transportation in a digital environment, is unnecessary according to the non-existence of physical laws.

Another common feature is the limited number of plots. The number of plots is limited since the earth's surface area is finite. IP2 argues that virtual real estate could similarly be limited and characterizes physical real estate as a rare commodity Chapter 2.3.1 and 2.3.2 mention in more detail the popular metaverse worlds Decentraland and Sandbox, both offering only a limited number of parcels, which makes virtual real estate rare and therefore more valuable.

5.5 Limitations and further research

This qualitative thesis study is to focus on stating the differences between physical and virtual real estate by the application of the concept of tokenization and its potential areas for future development and the challenges it faces in the current situation. The methodology used to gather data is based on theoretical concepts from the most recent literature reviews and information obtained from conducting a guided interview with professional experts in the field of this study's focus, which consists of six professional experts with years of experience in blockchain technology and one expert in the traditional real estate industry.

This master's thesis has various limitations. Currently, there are certain downsides to conducting expert interviews for data collection. No generalizations can be made because the seven experts were drawn from the German-speaking region, which was the focus of their perspective. The seven interviewed experts' knowledge and subjective experiences are the only foundation for the empirical findings.

Finding scholarly materials that link specific real estate market sectors to the blockchain is another constraint. Verifying and assessing as it should be in research is made difficult by employing only a small number of scientific sources concerning the application sectors. Additionally, this thesis's results and constraints provide potential approaches for exploration into this subject. However, neither in principle nor by the experts is a specific analysis of virtual real estate when discussing its potential and challenges.

Another limitation arises from the research question itself. The research question is an abstract question that gives the author a great deal of freedom in terms of the depth and breadth of the material to be

investigated. In the process, subject areas were weighted subjectively by the author, which led, for example, to the tokenization of physical real estate being discussed in greater detail in the expert interviews due to its complexity.

In this context, it would be worthwhile for future research to investigate the various aspects in more depth, as this work has established a basic foundation and thus serves as a first step. As regulatory challenges have emerged as one of the main reasons for the tokenization of physical real estate, it would be advisable for future expert interviews to be conducted with legal professionals who are specifically involved in this area, in order to better understand certain legislative thought processes.

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6 Conclusion

Blockchain technology is seen as a game-changer in many areas and industries. One application of this technology is tokenization, which has enormous potential, especially when it comes to tokenizing illiquid assets. A popular illiquid asset is real estate, which is classified as a very attractive and lucrative asset. However, due to high entry requirements and still-traditional nature of physical real estate transactions, relatively few individuals are able to invest in this asset class, creating various challenges for the real estate market. The blockchain, as a disruptive technology with the application of tokenization, offers solutions in this context. Not only is physical real estate considered a very attractive asset class for the portfolios of various investors, but also virtual real estate. Due to the hype around NFTs and the metaverse, virtual real estate is traded in digital environments, which could see enormous increases in value.

The objective of this thesis was to identify the differences in tokenization between physical real estate and virtual real estate, including a fundamental explanation of blockchain technology and its application in tokenization. Furthermore, potential and challenges were identified for physical and virtual real estate in relation to tokenization. The groundwork for achieving objectives was formed by a comprehensive review of the literature to define and describe in detail the technical concepts necessary for comprehension, and the observation results of expert interviews with seven professionals having extensive expertise in this field.

The empirical results provided a qualitative analysis of the differences between physical and virtual real estate in terms of tokenization. It was found that the tokenization process itself is very easy to implement on a technical level for both types, but when challenges and potentials are examined, enormous differences can be observed. While the tokenization of virtual real estate takes place entirely in a digital setting, the tokenization of physical real estate tries to connect the analog and digital worlds. The tokenization of physical real estate must first solve very complex challenges, such as regulatory problems and problems with the land register, so that the desired target state can be achieved. Once this state is attained, there will be enormous potential since physical real estate can then be immediately tokenized and traded on secondary markets. Eliminating complex processes and numerous intermediaries would improve the efficiency of this market. Fractional ownership could allow new business models to be established, such as rent to own, and real estate inheritances could be more easily resolved or divided. Current blockchain companies that deal with the tokenization of physical real estate currently only offer indirect investment opportunities. Before the necessary infrastructure for direct tokenization in this area is developed, it will probably take several years. In contrast to the tokenization of virtual real estate, the challenges relate to improving the current deployment. Therefore, the challenges deal not only with problems such as high gas fees and security but also with regulatory issues. Direct ownership and trading on secondary markets is already possible here, but the potential is still manageable. No one knows how the metaverse will develop and how virtual real estate will add value, but potential is already seen in connection with gaming and commercial nature.

Furthermore, the differences between physical real estate and virtual real estate, as its counterpart, in nature were examined. It was found that physical real estate has a completely different nature as it has the characteristics for an individual to live in it, thus fulfilling a basic need. And virtual real estate, on the other hand, is mainly bought for speculative purposes or as already mentioned for social activities. Other than said differences, similarities have been observed between the costs of physical and virtual real estate depending on their location. Thus, a property in a desirable location, such as a city center, makes it more expensive in both the physical and virtual real estate environment.

In summary, it can be stated that physical real estate and virtual real estate vary in many areas regarding tokenization, but also share similarities, based on the expert interview. Finally, it has been shown that tokenization in physical real estate provides an enormous potential that is considered to be much greater than its virtual counterpart after overcoming the challenges and developing the infrastructure. The only remaining uncertainty is when this will happen.

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List of abbreviations

DLT	Distributed Ledger Technology
ICO	Initial Coin Offering
IEO	Initial Exchange Offering
IP	Interview Partner
IPO	Initial Public Offering
FINMA	<i>Eidgenössische Finanzmarktaufsicht</i> (Swiss Financial Market Supervisory Authority)
FMA	Financial Market Authority
FTs	Fungible Tokens
M2M	Machine to Machine
MiCA	Markets in Crypto Asset regulation
NFTs	Non-Fungible Tokens
P2P	Peer to Peer
PoS	Proof of Stake
PoW	Proof of Work
SFTs	Semi-Fungible Tokens
STO	Security Token Offering
TCM	Token Container Model
TT	Trustworthy Technology
TVTG	<i>Token und VT-Dienstleister-Gesetz</i> (Tokens and TT Service Providers Law)
SPG	<i>Sorgfaltspflichtgesetz</i> (Due Diligence Act)
SPV	Special Purpose Vehicle

Annex

Interview Guide – Blockchain Experts (IP1, IP2, IP3, IP4, IP5, IP6)

General Questions

1. When did you first encounter blockchain technology and what fascinated you?
2. In which area do you see the greatest potential for blockchain technology?
3. What aspects of blockchain technology would you criticize?

Questions related to tokenization of physical real estate

4. How can I imagine how such tokenization of real assets works, according to your experience?
5. There have not been quite a lot of innovative processes/business models in the real estate market in the last decades. Do you think tokenization could revolute the real estate market?
 - 5.1 If yes, explain why?
 - 5.2 If no, why not?
6. According to your experience, where do you see the main challenges in tokenization related to real estate?
7. What do you think of the regulatory developments regarding tokenization for physical real estate?
8. Did the new blockchain laws bring any benefits to your company?
9. What potential does tokenization offer in terms of real estate?

Questions related to tokenization of virtual real estate

10. What do you think are the main reasons why someone would buy virtual land in an online environment?
11. Do you think that virtual real estate is just a bubble that is about to burst soon, or do you see any promising features/values in it?
12. What are the main differences between virtual and physical real estate? This question can be answered on different levels, such as technical level, regulatory level, and so on.

Interview Guide – Traditional real estate Expert (IP7)

Question about traditional real estate market

1. Can you tell me about the processes in a traditional real estate agency in your experience? What steps are involved?
2. What are the challenges in the traditional real estate market? Which processes are causing difficulties?

Question about Blockchain awareness

3. Have you ever heard of blockchain technology? What can you tell me about this technology?
4. The traditional real estate market has not really had much innovation in the last few years and decades. Would your company be open to incorporating a new technological solution, such as blockchain technology, into its operations?

Question about virtual real estate

5. What do you think are the main reasons why someone would buy virtual land in an online environment?
6. What do you think is the difference between physical real estate and virtual real estate?

Declaration of authorship

Differentiation of physical and virtual real estate in the application of tokenization

I hereby declare that the present paper is entirely my own work and without the use of any unauthorized assistance. Any content which has been taken verbatim or paraphrased from other sources has been identified as such. The insurance of independent work also applies to included drawings, sketches or graphic representations. This paper has not been submitted in any form whatsoever to an examining body nor has it been published. By submitting the electronic version of the final version of the thesis, I acknowledge that it can be checked for plagiarism with the help of a plagiarism detection service.

Balzers, 30/06/2022

Place, date

Damian Frick

First Name Last Name

A handwritten signature in black ink, appearing to read 'D. Frick', written in a cursive style.

Signature